Metallurgy

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Metallurgy

General Principles Process of Isolation of elements

Abundance of elements in the earth crust as:

O > Si > Al > Fe.....

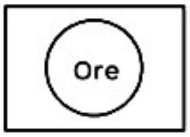
Non-metals Metals Transition metals
Elements are found in earth's crust in the form of minerals.
Insoluble compounds deposited on the earth crust.

Mineral :- Naturally occurring substance in which metal exists in free form or in combined form is called mineral.
Minerals are insoluble compounds.
Nitrates minerals does not exist because nitrates are water-soluble.
All minerals are useful
Mineral which are useful economical beneficial for metal extraction are called 'ore'

KEY NOTES

Assume we are taking a tour of the metallurgy industry. Easy understanding in case we accept all theoretical extractions as mentioned in the notes booklet. Let's talk about all the important points.

Ore :– Mineral from which metal can be conveniently & economically extracted is called ore.



Mineral ⊃ Ore

All ore are minerals but all minerals are not ores.

Gangue / Matrix :-
Small Impurities associated with ore are called matrix/gangue.

Points to remember

According to hard, soft acid-base theory, heavy metal (d and p- block belonging) will accept electrons from sulphides easily. So Sulphide are generally formed by metal ions.

Note

- Chalcogen are Group 16 element form ores because
- Most of the sulphides and oxides are water-insoluble.
- Polarizability of $O^{-2} < S^{-2}$ (Size of sulphur is greater).
- Polarizing power (d & p block) :
  Heavy metals > Lighter metals

Example: PbS Galena
$CuFeS_2$ copper pyrite
Ag_2S Argentite
HgS cinnabar
O^{-2}/ oxygen-containing ores are formed by lighter metal.

Example: $Al_2O_3.xH_2O \rightarrow$ bauxite

$CaCO_3 \rightarrow$ Limestone

G $_2$ $_3$ → Haematite

$CaCO_3.MgCO_3$ → Dolomite

d – block $\begin{cases} MnO_2 \rightarrow \text{Pyrolvsite} \\ FeCr_2O_3 \rightarrow \text{Chromite} \end{cases}$

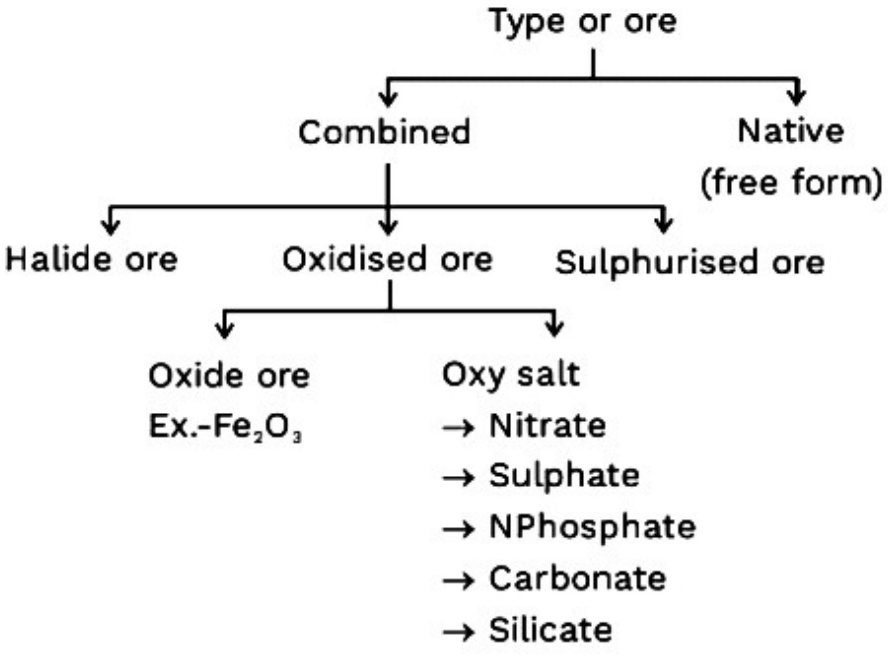
Classification of Ore

1. Native ore :

Found Only in free form
Example - Au, Pt, Ag, Hg
Free and combined form
Example -Ag & Hg

2. Combined form :

Except Au & Pt, all are found in combined form.



Different of Ores

OXIDE ORE

ZnO – Zincite

Fe_3O_4 – Magnetite

Cu_2O – Cuprite or Ruby copper

SnO_2 – Tinstone or Cassiterite

U_3O_8 – Pitch Blende

$AlO_x(OH)_{3-2x}$ – Bauxite (where $0 < x < 1$)

$Al_2(OH)_4Si_2O_5$ – Kaolinite (a form of clay)

$Fe.Cr_2O_4$ – $(FeO+Cr_2O_3)$Chromite ore

$Na_2B_4O_7.10H_2O$ – Borax or Tincal

$Ca_2B_6O_{11}.5H_2O$ – Colemanite

Fe_2O_3 – Hematite

$Fe_2O_3.3H_2O$ – Limonite

MnO_2 – Pyrolusite

TiO_2 – Rutile

$FeO.TiO_2$ – Ilmenite

SULPHURISED ORE :

Pbs Galena

ZnS – Zinc blende/sphalerite

HgS – Cinnabar

Cu_2S – Copper glance/Chalcocite

FeS_2 – Iron pyrite or Fool's gold

$CuFeS_2$ – Copper Pyrite (Chalcopyrite)

Ag_2S – Silver glance or Argentite

HALIDE ORE :

$NaCl$ – Rock Salt

KCl – Sylvite/silvine

Na_3AlF_6 – Cryolite

CaF_2 – Fluorspar

$AgCl$ – Horn Silver

$KCl.MgCl_2.6H_2O$ – Carnallite

Mixture of KCl and NaCl (sylvinite)

CARBONATE ORE :

$CaCO_3$ – Lime stone

$MgCO_3$ – Magnesite

$CaCO_3.MgCO_3$ – Dolomite

$FeCO_3$ – Siderite

$ZnCO_3$ – Calamine

$Cu)OH)_2.CuCO_3$ – Malachite or Basic Copper Carbonate

$Cu(OH)_2.2CuCO_3$ – Azurite

$PbCO_3$ – Cerussite

SULPHATE ORE :

$CaSO_4.2H_2O$ – Gypsum

$PbSO_4$ – Anglesite

$Na_2SO_4.10H_2O$ – Epsom Salt

$BaSO_4$ – Baryte

NITRATE ORE :

KnO_3 – Indian Salt petre

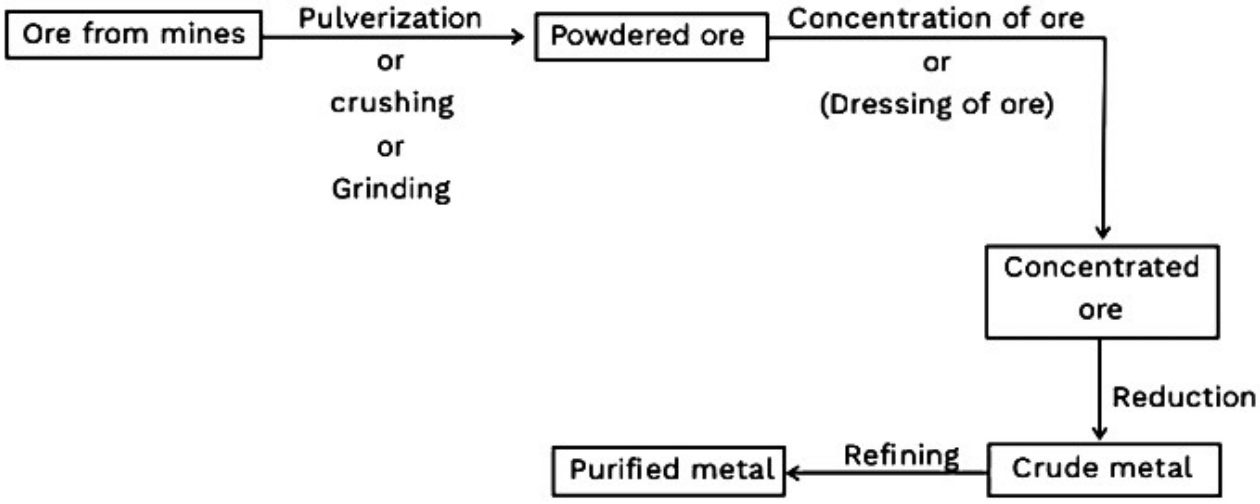
$NaNO_3$ – Chile Saltpeter

Metallurgy

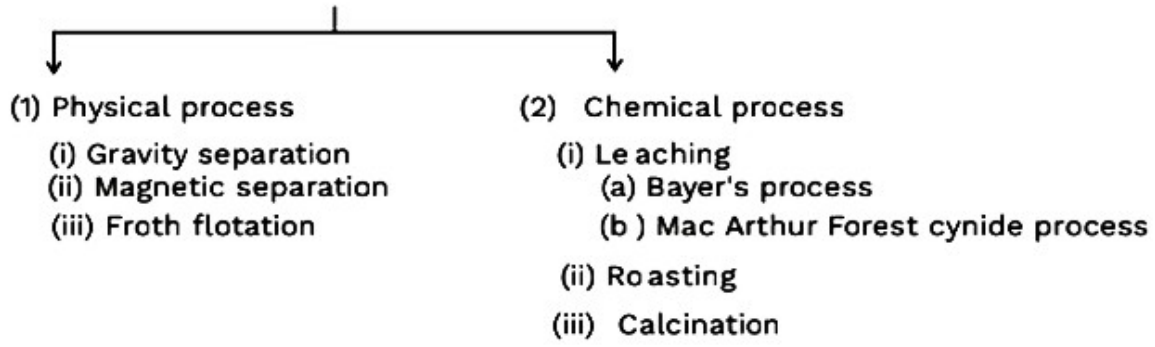
The processing metal from its ore is called metallurgy.

Extractive metallurgy: It is a Branch of metallurgy which deal with the extraction of metal from ore is called extractive metallurgy.

Extractive Metallurgy flow chart



The concentration of Ore/Dressing of ore/Beneficiation of ore:
The purpose of concentration is to remove visible impurities from ore to make it useful for further extraction.



Q1 Which of the following does not contain Mg :

(A) Magnetite (B) Magnesite (C) Asbestos (D) Carnallite

Sol. (A)

Q2 Which of the following is not an ore :

(A) Malachite (B) Calamine (C) Stellite (D) Cerussite

Sol. (C)

Q3 Carnallite does not contain

(A) K (B) Ca (C) Mg (D) Cl

Sol. (B)

Q4 Among the following statements, the incorrect one is :

(A) Calamine and siderite and carbonate ores

(B) Argentite and cuprite are oxide ores

(C) Zinc blende and pyrites are sulphide ores

(D) Malachite and azurite are ores of copper

Sol. (B)

Q5 Select the correct statement :

(A) Magnetite is an ore of manganese (B) Pyrolusite is an ore of lead

(C) Siderite is carbonate ore of iron (D) FeS_2 is rolled gold

Sol. (C)

Q6 "Fool's gold" is :
(A) Iron pyrites (B) Horn silver (C) Copper pyrites (D) Bronze

Sol. (A)

Q7 Assertion : Platinum and gold occur in native state in nature.
Reason : Platinum and gold are noble metals.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is not the correct explanation for statement-1.
(C) Statement-1 is true statement-2 is false
(D) Statement-1 is false, statement-2 is true.

Sol. (A)

Q8 Chloride ore among the following is :
(A) Malachite (B) Magnesite (C) Magnetite (D) Rock salt

Sol. (D)

Q9 A mineral is called an ore if :
(A) A metal can be economically obtained from it
(B) A metal can be extracted from it
(C) A metal can not be obtained from it
(D) It contains a precious metal

Sol. (A)

Q10 Corundum is :
(A) SiC (B) $Al_2^-O_3$ (C) $Al_2O_3.2H_2O$ (D) $KalSi_3O_8$

Sol. (B)

Q11 The rocky and silicious matter associated with an ore is called :
(A) Slag (B) Minera(C) Gangue (D) Flux

Sol. (C)

Q12 consider :
(a) Copper blende-Cu_2O (b) Pyrolusite = MnO_2
(c) Cryolite = Na_3AlF_6 (d) Cinnabar = PbS
Which is/are not correctly matched :
(A) a only (B) b only (C) d only (D) a and d

Sol. (D)

Q13 Bauxite has the formula :
(A) $Al_2O_3.H_2O$ (B) $Al_2O_3.2H_2O$ (C) $Al_2O_3.3H_2O$ (D) $KAlSi_3O_8$

Sol. (B)

Q14 The most common ore of copper is :
(A) Cu_2S (B) $CuFeS_2$ (C) $CuCO_3.Cu(OH)_2$ (D) Cu_5FeS_4

Sol. (B)

Q15 Calamine is an ore of :
(A) Zn (B) Mg (C) Ca (D) Pb

Sol. (A)

Concentration of Ore

The process of removing matrix or gauge from powdered/pulverized ore with the help of the following methods.

1. Physical process
2. Chemical process

1. Physical processes :

Based on some physical properties of metal and ore, these processes classified into three different methods

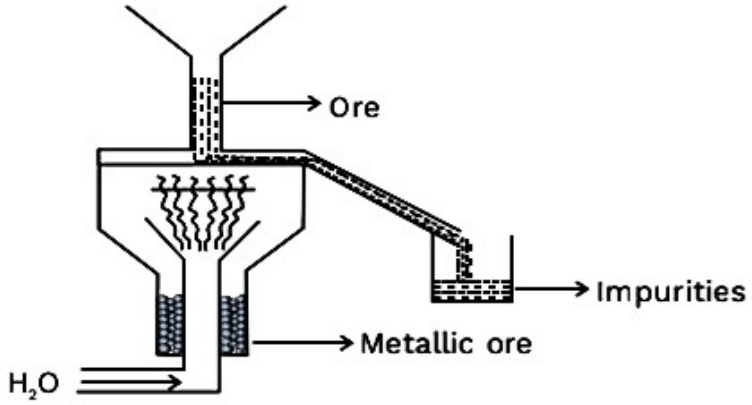
I. **Gravity Separation / Levigation:**

It is a method to remove some light and small impurities with the help of some particular types of equipment.

(a) Hydraulic washing

When the ore is heavy compared to impurities, then ore is concentrated by Gravity separation, which is based on Differences in densities.

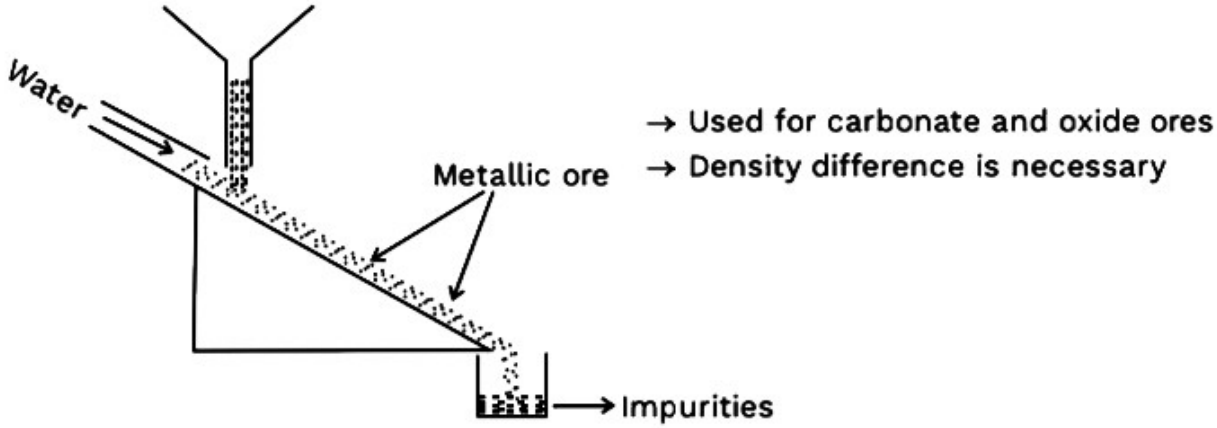
Concept - ($d_{Ore} > d_{impurities}$)



When powdered ore is mixed in running water, heavy ore particles settle down while impurities are flushed away with water.

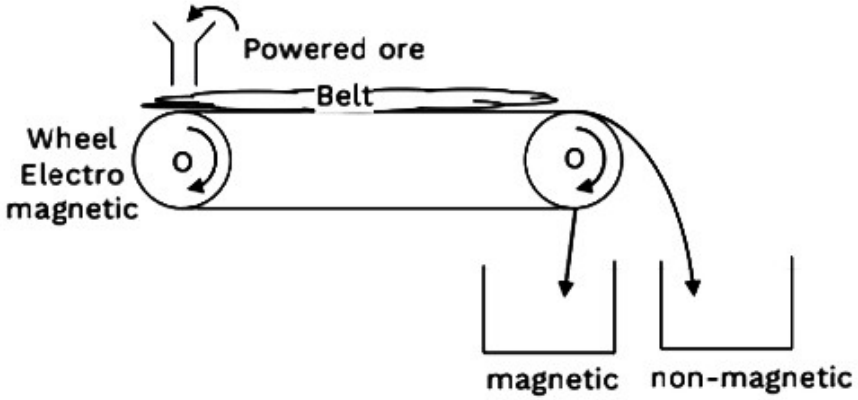
Note

This process is also done by Wilfley table.



II. **Magnetic Separation:**

When ore of the component in powdered ore, either ore or impurities are magnetic, and the other is non-magnetic, then that are separated by magnetic separation.



Example:

Magnetic		Non-magnetic	
Fe_2O_3, MnO_2, $FeCr_2O_4$	Ore	SiO_2	impurity
$(FeWO_4 + M_nWO_4)$	Wolframite	impurity	
(SnO_2)	Ore		

III. **Froth Floatation:**

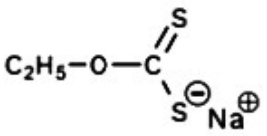
This process is used for sulphide ore because Sulphide ore is formed by heavy metals having high polarising power.

Concept ⇒ In this process, the following concept are responsible

(a) Adsorption (surface attraction) – For example, in Heber's process, Fe is used as a catalyst so that hydrogen gas does synergic bonding on the catalyst's surface, which is adsorption.

(b) Like dissolve Like – It means polar solvent dissolves polar substance and non-polar dissolve non-polar substance.

(c) Differential wet ability – To convert sulphide ore into non-polar by adding xanthate because xanthate having two different polar and non-polar parts.

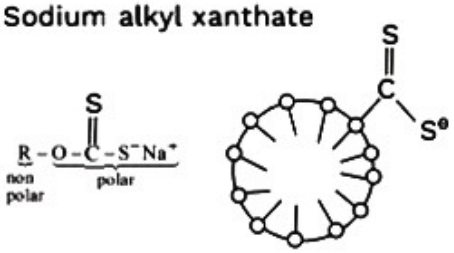


Non-polar polar

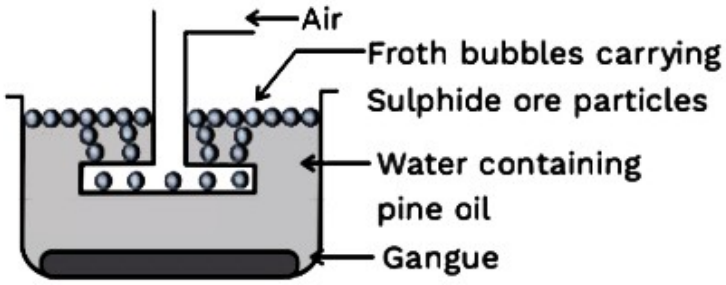
Process:

Ore particles are ionic compounds to convert them into non-polar species, and xanthate is used. Xanthate has two parts. One is polar (hydrophilic), which attracts high polarised sulpure ore particles. Xanthate is called a collector because they are used to collect particles.
Example: Sodium alkyl xanthate

Sodium alkyl xanthate



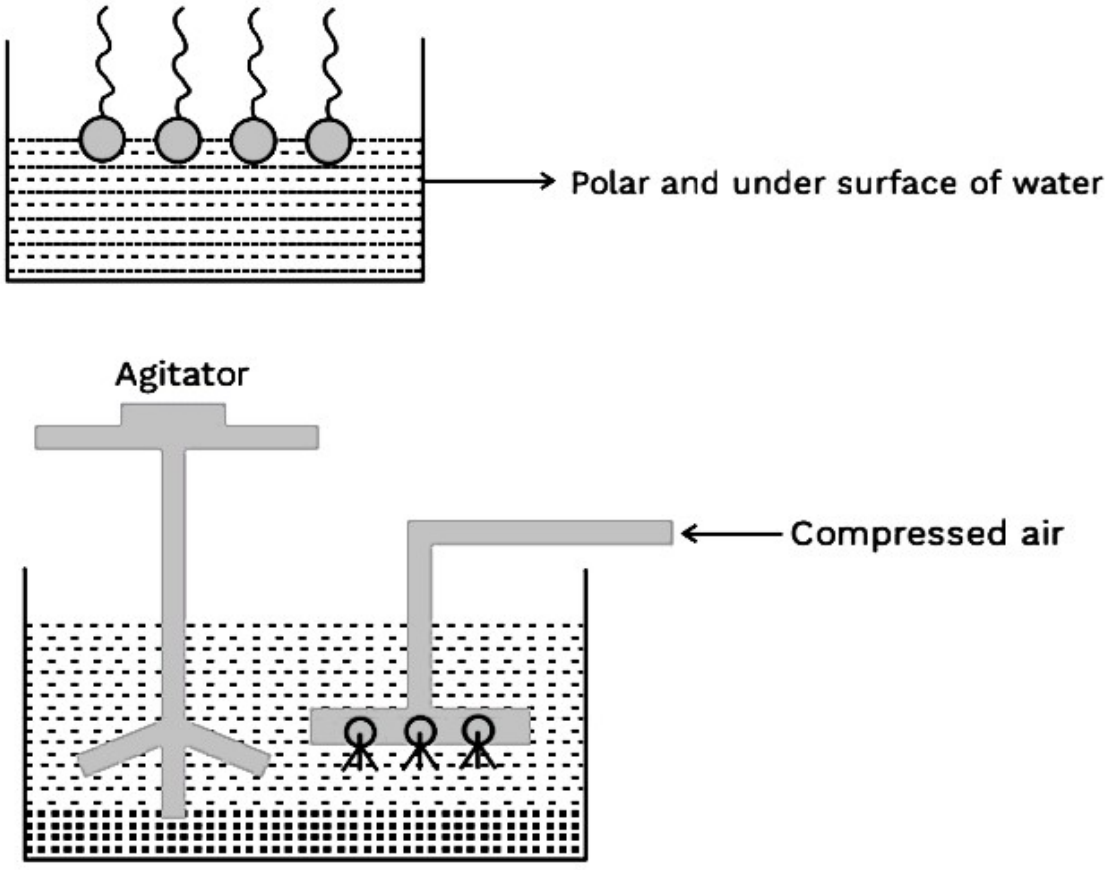
Ore particles are wetted with pine oil, while impurities (gangue) are wetted by water because they are heavy and settle down in a container.



Pine oil acts as frother as well as collector

Cresol and Aniline are used as froth stabilizer because the decreases surface tension of water.

Froth flotation process diagram



(Water + Pine oil + Metallic ore + Sodium ethyl xanthate)

Ore is recovered by froth by

(i) Heating –It is due to break surface attraction (i.e. Adsorption)

(ii) Using dilute HCl (strong Acid) -Xanthate is an anion of weak Acids, and they are a strong base with a high affinity for H^+ so ore will remove easily.

Note

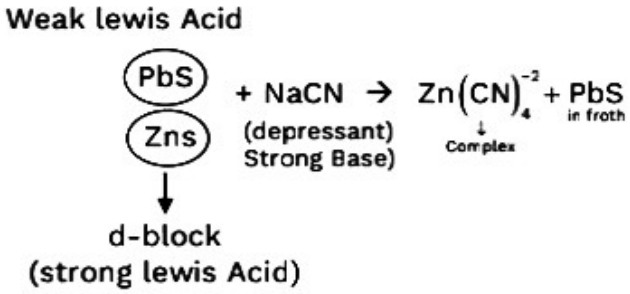
We also use depressants and activators in the froth flotation mixture, which is explained in the next segment.

Selective Froth Flotation :

In this method, a chemical is added in the froth flotation process that makes the process selective. It depresses the movement of other metal sulphide particles present as impurities. This is known as selective froth floatation.

Pbs (Galena) contain impurity of Zns, so NaCN is used as a depressant.

P-block



Recovery of Zn-

To recovery of Zn we use the $CuSO_4$ as a activator, and the following reaction takes place

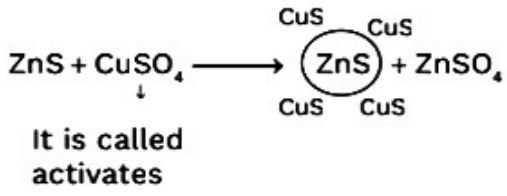
$$S^{-2} + Zn(CN)_4^{-2} + CuSO_4 \rightarrow \underset{\text{Cyanogen gas}}{(CN)_2 \uparrow} + \left[Cu(CN)_4\right]^{-3} + ZnS$$

Due to Solubility of CuS < ZnS, because

$Cu^{+2} \Rightarrow d^9$, adsorption increase due to High polarising power.

$Zn^{+2} \Rightarrow d^{10}$, adsorption decreases due to less polarising power.

So to Enhance the adsorption of Zn, $CuSO_4$ is used



So this is a complete froth flotation method for the concentration of ore.

Q16 $Ag_2S + NaCN + Zn \rightarrow Ag$

This method of extraction of Ag by the complex formation and then its displacement is called

(A) Part's method **(B) Mac-Arthur, Forrest method**

(C) Serpek method **(D) Hall's method**

Sol. (B)

Q17 Which ore of the following is not a method of concentration of ore?
(A) gravity separation (B) froth floating process
(C) electromagnetic separation (D) smelting

Sol. (D)

Q18 Chemical leaching is useful in the concentration of
(A) Copper pyrites (B) bauxite (C) galena (D) cassiterite

Sol. (B)

Q19 In froth-floatation, process, pine oil function as
(A) Activator (B) Frother (C) Collector (D) Agitator

Sol. (B, C)

Q20 In the cyanide process involving the extraction of silver, zinc is used industrially as a (an).
(A) oxidising agent (B) reducing agent (C) solvent (D) solvating agent

Sol. (B)

Q21 During initial treatment, Preferential wetting of ore by oil and gangue by water takes place in-
(A) Levigation (B) Froth flotation
(C) Leaching (D) Bessermisation

Sol. (B)

Q22 The Mac-Arthur Forrest process is used for the extraction of metal

(A) Zn (B) Cu (C) Fe (D) Ag

Sol. (D)

Q23 In froth floatation, the oil used is

(A) olive oil (B) Pine oil (C) Coconut oil (D) Mustard oil

Sol. (B)

Q24 Bayer's process is used for the purification of bauxite containing ..... as impurity

(A) silica (B) rutile (C) Fe_2O_3 (D) None

Sol. (C)

Q25 An ore of tin containing $FeCrO_4$ is concentrated by

(A) Magnetic separation (B) Froth floatation

(C) Electrostatic (D) Gravity separation

Sol. (A)

Q26 Magnetic separation process used for the concentration of

(A) chalcopyrite (B) bauxite (C) hematite (D) colamine

Sol. (C)

Q27 Which of the ore dressing process require the finest size of ore?

(A) froth floatation (B) gravity separation

(C) magnetic separation (D) All

Sol. (D)

Q28 **Which of the following ores is concentrated by leaching?**
(A) pyrite **(B) galena** **(C) epsomite** **(D) argentite**

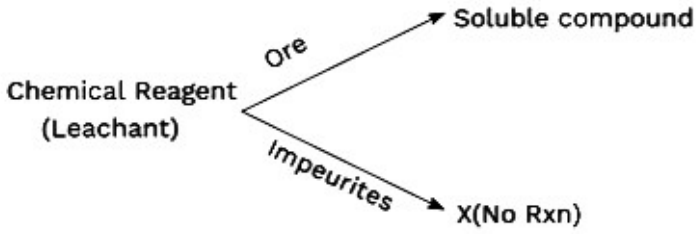
Sol. **(D)**

Q29 **Leaching is a process of**
(A) peduction **(B) concontration** **(C) refining** **(D) oxidation**

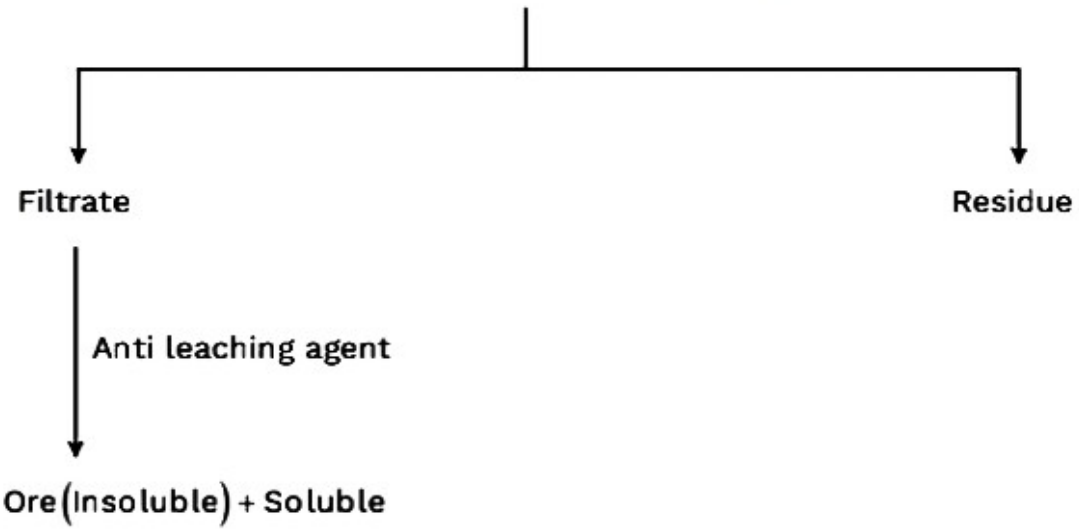
Sol. **(B)**

2. Chemical processes

1. **Leaching method:** Process of selective dissolution of ore in the concentrated aqueous solution of some chemical called a leaching agent at room temperature.



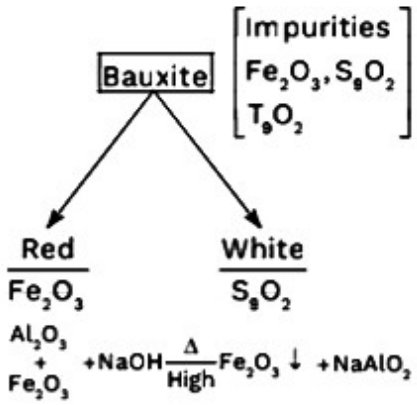
Now apply a filter so we will get



Types of leaching

(i) Bayer's process

- The process is used for bauxite $Al_2O_3, 2H_2O$



Considered High temperature for more solubility.

After filtration :

$$NaAlO_2 \xrightarrow[\text{TemLow}]{(i)CO_2} Na_2CO_3\,(aq) + Al(OH)_3 \downarrow$$

$[Na_2O.Al_2O_3]$

CO_2 is weak acidic favour reaction with Na_2O but not with $Al(OH)_3$.

Points to remember

$NaOH(Base) + Fe_2O_3(Base) \rightarrow X$

$Al_2O_3(amphoteric) + NaOH(Base)$
$\rightarrow NaAlO_2(Soluble)$

(ii) Mac Arthur Forest cynide process :

This process is used for native Ag, Au and Ag_2S.

Leaching agent : NaCN and O_2

Anti leaching agent used as Zn powder.

For Ag/Au

$$M\ \{Ag/Au\} + O_2 + NaCN \rightarrow \underset{\text{Scluble}}{Na[M(CN)_2]} + \underset{\text{insoluble}}{\text{impurities}}$$

After filtration

$$Na[M(CN)_2] + Zn \rightarrow \underset{\text{inscluble}}{M} + \underset{\text{inscluble}}{Na[Zn(CN)_2]}$$

For Ag_2S

$$Ag_2S + NaCN \leftrightarrow Na[Ag(CN)_2] + S^{-2}$$

$$Ag_2S + NaCN + O_2 \rightarrow Na[Ag(CN)_2] + S\downarrow + SO_4^{-2}$$

$$Na[Ag(CN)_2] + Zn \rightarrow Na[Zn(CN)_2] + Ag\downarrow$$

Points to remember

Ksp of Ag_2S is very low so first reaction does not move forward direction so air is used to oxidised S^{-2} in to S/ SO_2 / SO_4^{-2}.

Pay attention !!

We are going to reduce ore, and generally, carbon(coke powder) is used for the reduction of ore because carbon is cheap, and as temperature increases, the oxygen affinity of carbon is also increased in comparison to all elements. So carbon is the most important element used for reduction. But the question is arises!!

Is reduction by heating with carbon is valid for all?

The answer is No.

For understanding this whole concept, we need to understand extractive metallurgy first.

Types of Extractive metallurgy

Reduction of metal is an important process of extractive metallurgy, so based on reduction, metallurgy can be divided

(a) Electra metallurgy ⇒ Reduction by using electricity

(b) Hydrometallurgy ⇒ Reduction by using water

(c) Pyrometallurgy ⇒ Reduction by using heat

Tips and Trick

If the leaching agent is acid use a basic anti-leaching agent and vice versa.

Reactivity Series:

Highly reactive metals (Electropositive metals)

Li
K
Ba
Sr
Ca
Na
mg
Al

} Electro metallurgy (s – block to Al)

Moderate reactivity metals

Mn
Zn
Cr
Fe
Cd
Co
Ni } Pyro metallurgy
Sn
Pb
H_2
Cu
Hg

Low Reactive metal

Ag
Au } ⇒ Hydro metallurgy used
Pt

So we can clearly understand that Highly electropositive metals will not reduce by heating (pyro) with Carbon, they reduced by electrical methods; similarly, the less reactive metals will not be reduced by Heating. Because they are expensive and generally extracted by hydrometallurgy. So moderately reactive metals are suitable to reduce by heating (pyro) methods.

Key concept

For Ag/Au/Pt and low-grade Cu (Ore in which copper contains is very little), we use hydro-metallurgy along with iron scrape or bacteria. In this process, hydrogen gas will form, and this hydrogen gas is responsible for the reduction of metals like low-grade copper.

Now we know the concept of pyrometallurgy, and pyrometallurgy is easy for oxides because the reduction of sulphide is non-spontaneous, so before reduction, the ore is converted into oxide

Example:

$$ZnS + C \longrightarrow ZnO + CS_2 \qquad \Delta G = +ve$$

$$\downarrow O_2$$

$$ZnO + CS + O_2 \longrightarrow Zn + CO_2 \qquad \Delta G = -ve$$

Explanation:

In case of reduction of sulphide the CS_2 gas will produce and due to 2p-3s overlapping between C=S bond, strength is weak and the process is non-spontaneous while in case reduction of oxide, CO_2 will produce and due to 2p-2p overlapping the bond between C=O is stronger, and the process is spontaneous.

So, conversation step is an important part of pyrometallurgy, and it is done by Chemical processes like Roasting and Calcination.

1. **Roasting**	2. **Calcination**
Heating of ore in the presence of air below the melting point. e.g. $ZnS + O_2 \rightarrow ZnO + SO_2$ Benefits and cause of Roasting • Ore become porous • Surface area increases • Oxidation Reaction • Exothermic Reaction • Used for sulphide ore. • Volatile impurities SO_2, Al_2O_3, Sb_2O_3 • Organic impurities ↓ decompose $ZnS + O_2 \rightarrow ZnO + SO_2$	Heating of an ore in absence of air below the melting point. e.g. $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2 \uparrow$ Benefits and cause of Calcination • Ore become parous • Surface area increases • Thermal decomposition • Endothermic reaction • Used for hydroxide and oxo salts • Volatile impurities $H_2O, CO_2, \underbrace{S, As, Sb}_{\text{elementary form}}$ • Organic impurities ↓ decompose $ZnCO_3 \rightarrow ZnO + CO_2$

The Roasting is also done in the presence of a limited supply of oxygen which is important to understand

Partial Roasting / Controlled roasting :
Heating of ore in the presence of a limited supply of oxygen and is done for selective oxidation and self-reduction.

(a) Selective oxidation :

In this method, copper pyrite ($CuFeS_2$) is heated with limited oxygen .

Explanation: Iron reactivity is grater then copper; similarly oxygen affinity is greater for iron in comparison of copper, so whenever we supply limited oxygen, Fe react first with oxygen and form iron oxide (even some amount of copper oxide also form and further convert into iron oxide) and This iron oxide impurity remove by adding SiO_2 (known as Flux) in the form of Slag ($FeSiO_3$). CuS_2 form in this method is known as copper matte.

The reaction involves in Roasting

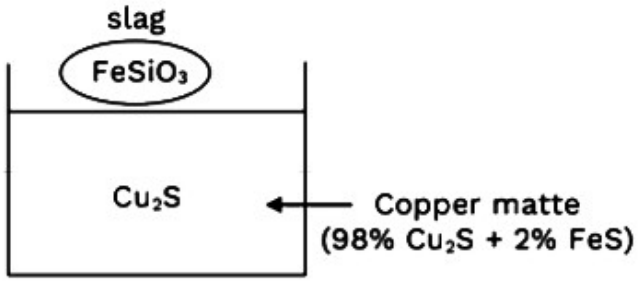
$$CuFeS_2 + \text{limited } O_2 \xrightarrow{\Delta} FeO + Cu_2S + SO_2 \uparrow$$

$$(Cu_2S + FeS + FeS_2)$$

$$\left[\begin{array}{l} Cu_2S + O_2 \rightarrow Cu_2O + SO_2 \\ Cu_2O + FeS \rightarrow FeO + Cu_2S \end{array}\right]$$

$$FeO \xrightarrow[\text{flux}]{SiO_2} FeSiO_3$$

Slag ⇒ (lighter, less dense, Float on the surface)



Slag ⇒ slag are immiscible compounds, less dense, float on the surface of molten metal. When basic oxide impurity reacts with acidic oxide, ore acidic oxide impurity reacts with basic oxide they form slag.

Slag formation is an acid-base reaction in a molten state

$$\text{(Basic) } FeO + \overset{\text{Acidic flux}}{SiO_2} \longrightarrow FeSiO_3$$

$$\text{(Acidic) } SiO_2 + \underset{\text{Basic flux}}{CaO} \longrightarrow CaSiO_3$$

Flux ⇒ flux are oxides, used to remove oxide impurities from slag.

Slag is a thermally stable compound.

e.g. Silicate, Phosphate

Silicate slag ⇒ used in cement industry

Phosphate Slag $\Rightarrow$ used in the fertilizer industry
Thomas Slag $\Rightarrow$ $CO_3(PO_4)_2$

Other slags are not possible due to thermal stability.

(b) Self-reduction :

The process of self-reduction is the reduction of oxide ore which is made by partial roasting.

Explanation:

After a certain temperature, the Oxygen affinity of of sulphure is less then Cu, Pb, Hg ($S > Cu / Pb / Hg$), So sulphide ore of Cu/Pb/Hg are reduced by partially roasted ore. This is called self-reduction.

Reaction involves in Self reduction process

$$PbS + \underset{\text{(limited)}}{O_2} \rightarrow PbO + SO_2 \uparrow$$

$$PbS + PbO \rightarrow Pb + SO_2 \uparrow$$

Q30 **Heating of carbonates to remove carbon is called as**

(A) Roasting **(B) Calcination** **(C) Smelting** **(D) Fluxing**

Sol. **B**

Q31 **Which one of the following reactions is example of the calcination process**

(A) $2Ag + 2HCl + (O) \longrightarrow 2AgCl + H_2O$ **(B)** $2Zn + O_2 \longrightarrow 2ZnO$

(C) $2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$ **(D)** $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

Sol. **D**

Q32 **The main function of roasting is:**

(A) To remove the volatile impurities **(B) oxidation**

(C) reduction **(D) To make slag**

Sol. **B**

Q33 Roasting and calcination is not an essential step for extraction metal from this ore-

(A) Chalcocite (B) Galena (C) Argentite (D) Cassiterite

Sol. C

Q34 Which of the following reaction represent a calcination reaction?

(A) $HgS + O_2 \longrightarrow Hg + SO_2$

(B) $AgNO_2 + NaCl \longrightarrow AgCl + NaNO_3$

(C) $CuCO_3 \cdot Cu(OH)_2 \longrightarrow CuO + CO_2 + H_2O$

(D) $Al_2O_3 + NaOH \longrightarrow NaAlO_2 + H_2O$

Sol. C

Q35 Roasting is used during metallurgical operation using:

(A) Galena (B) Iron pyrites (C) Gopper glance (D) All

Sol. D

Q36 PbS is roasted to give

(A) PbO_2, SO_2 (B) $PbO, PbSO_4$ (C) $PbO_2, PbSO_4$ (D) $PbO, PbSO_4, SO_2$

Sol. D

$$PbS \xrightarrow{\Delta} PbO + SO_2$$
$$\downarrow O_2$$
$$PbSO_4$$

Q37 Heating of pyaites to remove sulphur is called as

(A) roasting (B) calcination (C) smelting (D) fluxing

Sol. A

Q38 In the extraction of lead by self-reduction process, the reducing agent is-

(A) PbS (B) O_2 (C) C (D) Al

Sol. A

Q39 Roasting is carried out in

(A) blast furnace (B) Reverberatery furnace

(C) A muffle furnace (D) An electric furnace

Sol. B

Q40 Which of the following is not a basic flux

(A) $CaCO_3$ (B) CaO (C) SiO_2 (D) MgO

Sol. C

Q41 Sintering process is, in which

(A) Roasting of ore takes place (B) calcination of ore takes place

(C) reduction of ore takes place (D) melting of ore takes place

Sol. D

THERMODYNAMICS OF REDUCTION PROCESSES (ELLINGHAM DIAGRAM)

This diagram explains the selection of reducing agents for metal oxide at a suitable temperature.
Ellingham plotted the curve between ΔG and Temperature for metal oxide formation.

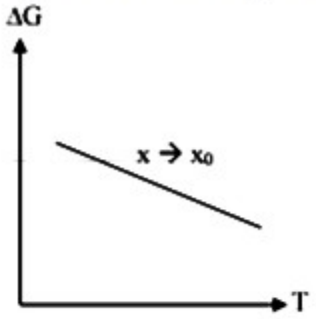
Concept: We know that for a spontaneous reaction, the free energy ΔG must be negative. He observed when metal react with oxygen and form oxide

$$M + O_2 \longrightarrow MO$$

So, according to the relation between ΔG and ΔS
$\Delta G = \Delta H - T\Delta S$ (compare above equation with straight-line equation), Graph ΔG v/s Temperature found to be graph is a straight line. The following observation for this graph is given below

(I) If ΔS = +ve, slope = –ve

As temperature increases
ΔG decreases
Spontaneity increases
Oxygen affinity of x increases
It means product formation is easy, or we can say Metal oxide formation is easy.



(II) If ΔS = -ve, slope = +ve

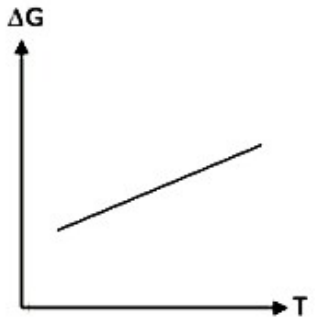
As temperature increases
ΔG increases.
Spontaneity decreases
Oxygen affinity of x decreases
It means product formation is not easy, or we can say Metal oxide does not form quickly.



Different-Different graph for a different condition

1. **Formation of metal oxide:**

   $M_{(s/\ell)} + O_{2(g)} \longrightarrow MO_{(s)}$; Δs = –ve, slope = +ve When the above reaction solid/liquid metal reacts with gaseous oxygen, it will form solid metal oxide so that entropy decreases and entropy change negative, and slope will be positive.



So as temperature increases, the oxide formation tendency of metal decreases.

2. **Formation of metal oxide from the different-different state of metal:**

   ΔS

   (i) When metal is gaseous

   $M_{(g)} + O_{2(g)} \longrightarrow MO_{(s)}$ ΔS_1 = Negative

   (ii) When metal is liquid

   $M_{(l)} + O_{2(g)} \longrightarrow MO_{(s)}$ ΔS_2 = Negative

   (iii) When metal is solid

   $M_{(s)} + O_{2(g)} \longrightarrow MO_{(s)}$ ΔS_3 = Negative

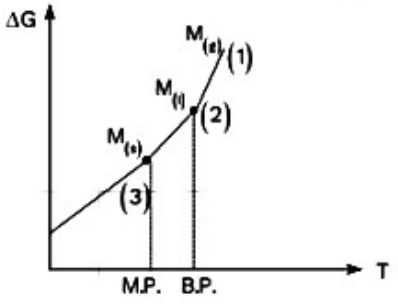
   Above reaction, the entropy changes are negative, so comparison of Negative value of entropy change as $\Delta S_1 > \Delta S_2 > \Delta S_3$

   [due to entropy of g > l > s]

   So Slope : $\Delta S_1 > \Delta S_2 > \Delta S_3$

   [Equation (i) > Equation (ii) > Equation (iii)]

   By ploting graph following observation takes place as given below in the graph



In above graph the Entropy of reactant (1) > (2) > (3)
(due to entropy of g > l > s)

And Entropy of product (1) = (2) = (3)

So Change in entropy ΔS = (product – Reactant)

$$\Delta S = (3) > (2) > (1)$$

Slope = (1) > (2) > (3)

Observation: In the above graph (1), continuously going upward so that ΔG increases and spontaneity decreases.

Conclusion: By the above observation, As ΔG increases and spontaneity decreases, Metal oxide reduction is easy when metal is present in the gaseous state, but as required temperature is high in the gaseous state so basically reducing agent has taken accordingly, and Ellingham diagram tells us that for which temperature range at which metal reduce in the presence of a suitable reducing agent.

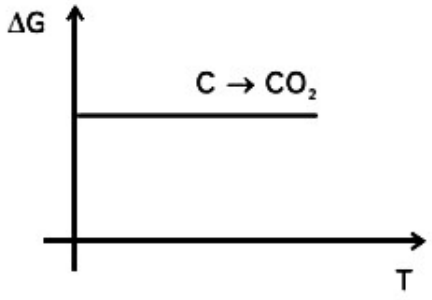
We know that carbon is a reducing agent, and carbon forms a different oxide at different temperature ranges, so let's talk about observation for different oxides when reduction is easy.

3. For Carbon dioxide (CO_2) :

The reaction of 1-mole carbon with 1-mole oxygen, 1-mole carbon dioxide will form.

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \; ; \Delta S = 0$$

1 mole 1 mole



Observation: In the above graph, as temperature increases, the carbon affinity remains constant, which means after a specific temperature, carbon reduces metal oxides, because as we have seen that metal's oxygen affinity decreases.

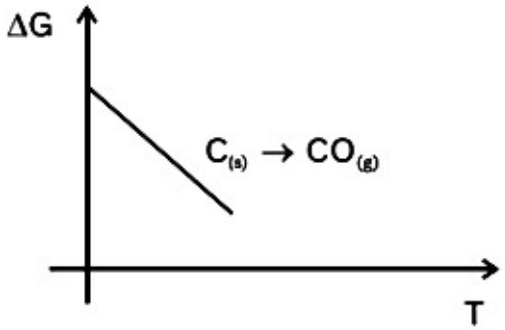
4. **For carbon mono oxide(CO) :**
   The reaction of 2-mole carbon with 1-mole oxygen forms 2-mole of carbon monoxide.

$$2C_{(s)} + O_{2(g)} \longrightarrow 2CO_{(g)} ;$$

1 mole 2 mole

ΔS = +ve

Slope = –ve



Observation: In the above graph, as temperature increases, the carbon affinity increases and metal oxygen affinity decreases so it is more convenient method form above but this is in presence of high temperature zone so it is expensive method.

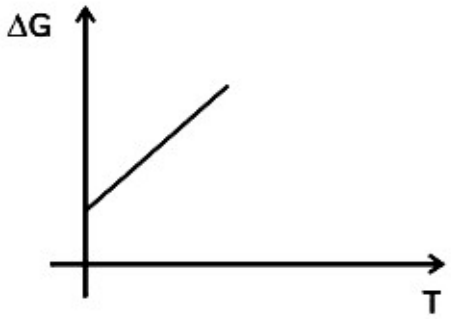
5. **For 2 mole CO_2 from carbon monoxide:**

$$2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)} ;$$

2 mole 1 mole 2 mole

ΔS = -ve

Slope = +ve

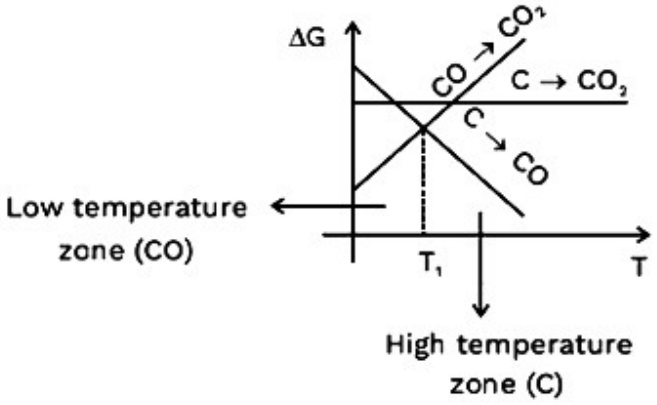


Observation: In the above graph, as temperature increases, the carbon mono oxide converts into carbon dioxide decreases as temperature increases.

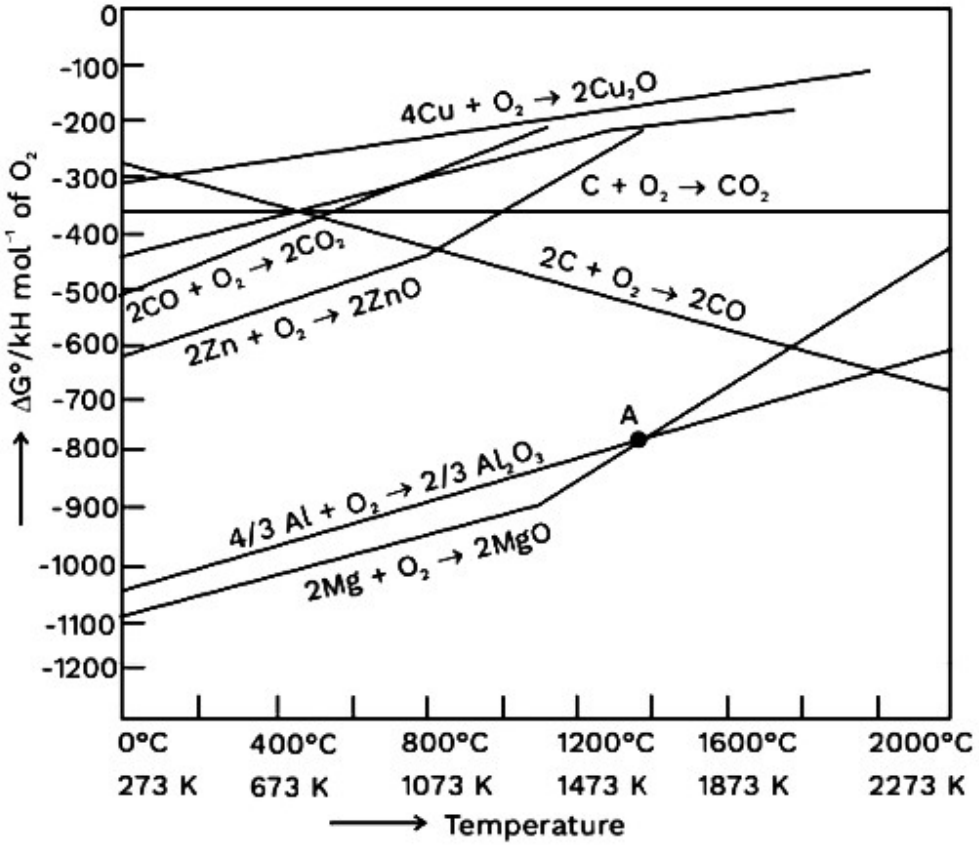
Conclusion: By above (3),(4) and (5) we found that formation of carbon dioxide by carbon mono oxide is more exotherming so we can say that in low temperature zone carbon monoxide (CO) is a good reducing agent due to less (negative) ΔG and at high temperature zone Carbon (C) is a good reducing agent due to less(negative) ΔG. Explained with below graph

	ΔH	
$C \longrightarrow CO$	Endothermic	...(1)
$C \longrightarrow CO_2$	Exothermic	...(2)
$CO \longrightarrow CO_2$	More exothermic	...(3)

ΔH = (1) > (2) > (3)



Explanation of Ellingham:



As temperature increases, oxygen affinity of metal decreases but oxygen affinity of carbon increases so carbon can act as a reducing agent for all metal oxide due to low cost, high oxygen affinity but :

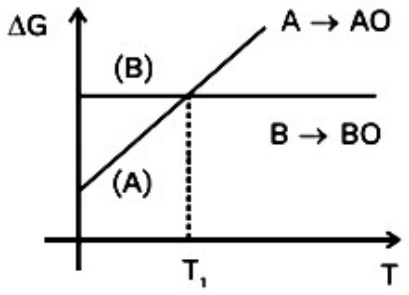
(i) For s-block and aluminium required temperature for carbon reduction is very high, and they also form carbide, so they are reduced by electrolysis of molten salt.

(ii) Ag, Au, Pt oxides are unstable, so carbon reduction is not useful, so due to low reactivity, they can be easily reduced by the metal displacement method.

(iii) After a certain temperature oxygen affinity of sulphour is greater than Cu, Pb and Hg, their partially roasted sulphide ore is reduced by self-reduction process.

(iv) MnO_2 and Cr_2O_3 due to high lattice energy, high activation energy can't reduce by carbon. They are reduced by aluminium (Alumino thermite process) because the formation of Al_2O_3 can provide competent activation energy.

Example :



$\Delta G \downarrow$ forward $\downarrow$

So, metal oxides for easily

For (A) $\Delta G \downarrow$ reaction forward, so for (A) oxygen affinity increases.

For (B) $\Delta G \uparrow$ reaction forward but less forward as compared to (A) so oxygen affinity less as compared to (A)

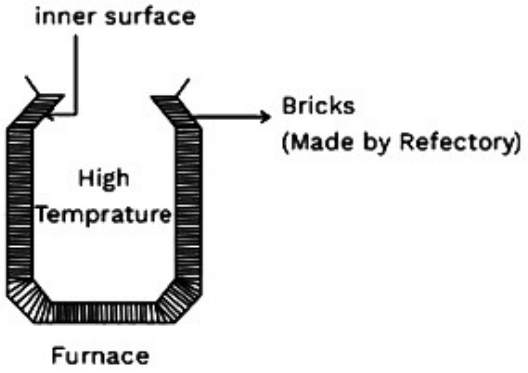
Refractory:

Refractory are heat absorbers used to absorb heat in the furnace for the extraction of metal.

For Refractory, High Melting point Compounds are used in the form of bricks.

For Example$\Rightarrow$ CaO, MgO, SiO_2 (lattice energy is high)

They Protect the inner/outer surface of the furnace from High temperature.



Reduction of Ores :

In the Process of extraction of metal, concentrated ore moves for Reduction, and different metal ores different
reduction methods are there, but first, one term is frequently used in the Reduction i.e. Smelting, let's understand
the Smelting.

Smelting : Smelting means slag formation or Carbon Reduction.

Note

We already learned about Roasting and Calcination, but in both processes, a condition is necessary to heat ore below its melting point, but in the case of Reduction, we will melt metal in every situation.[Slag floats on molten metal, so melting is required.]

Q42 **The Common method of extraction of metal from oxide ores;**
(A) reduction with carbon **(B) reduction with Hydrogen**
(C) reduction with Aluminium **(D) Electrolytic method**

Sol. **(A)**

Q43 **Which of the following metals cannot be extracted by carbon reduction.**
(A) Pb **(B) Al** **(C) Hg** **(D) Zn**

Sol. **(B)**

Q44 **Purpose of Smelting of an Ore is-**
(A) To oxidize it
(B) To reduce it
(C) Tom remove vapourizable impurities
(D) To obtain an alloy

Sol. **(B)**

1. **Reduction of iron :**

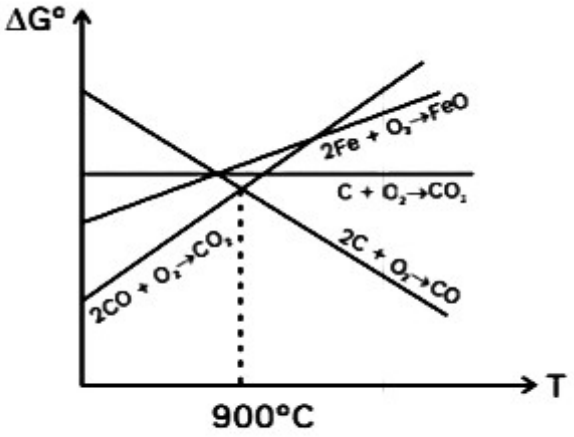
Ore of iron ⇒ Fe_2O_3 (Hematite) [SiO_2 present as a impurity]

Flux used ⇒ $CaCO_3$

Reducing agent used ⇒ Carbon (C)

We already learn in the Ellingham diagram that at low-temperature zone carbon mono oxide and at high temperature zone, carbon act as reducing good agent.

In the given curve, it is clear that for iron, below 900°C, carbon mono oxide use for Reduction, while Above 900°C, carbon will use for the Reduction of iron.



Reduction takes place in the Blast furnace, which is approx 100 feet long

Flow chart

Put the raw material (ore + limestone + coke) by crane from Throat of furnace

⇓

combustion Zone

- combustion (due to Hot air)
- $C + O_2 \longrightarrow CO_2 + Energy$
- ⇓
- $CO_2 \longrightarrow CO$(By increasing Temperature)
- ⇓
- Carbon monoxide move upward

⇓

Fe_2O_3 move downward.

⇓

Reduction Zone at 900°C

$Fe_2O_3 \longrightarrow Fe_3O_4 \longrightarrow FeO \longrightarrow Fe_{(s)}$

$CO \longrightarrow CO_2$

(Fe_2O_3 can be reduced before 900°C)

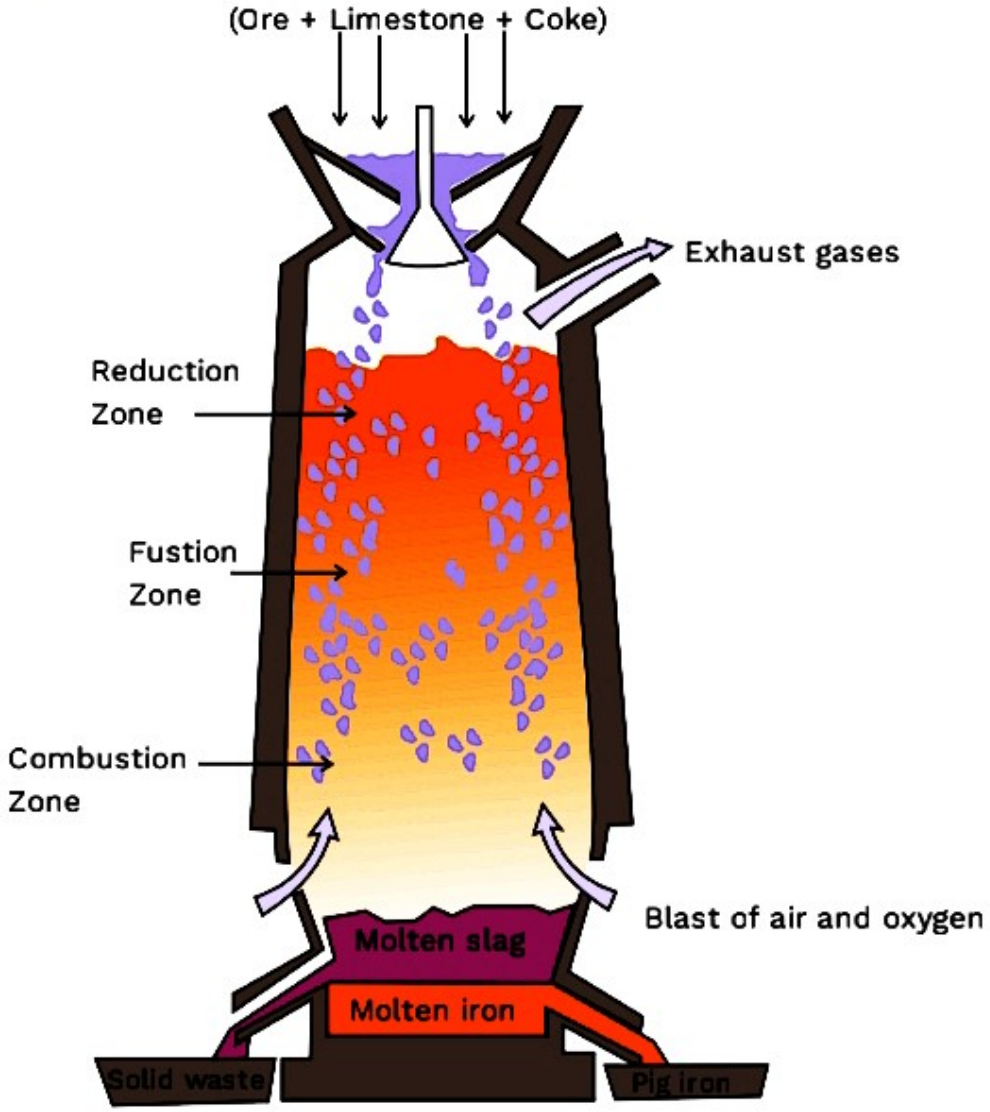
$\Downarrow$

Slag formation $\begin{bmatrix} CaCO_3 \longrightarrow CaO \\ CaO + SiO_2 \longrightarrow CaSiO_3 \end{bmatrix}$

$\Downarrow$ ($CaSiO_3$ used in cement Factory)

Fusion zone $\left[Fe_{(s)} \longrightarrow Fe(\ell)\right]$

The iron we get is called as **pig iron.**



Blast Furnace

Working :
The blast furnace is charged by raw material (Fe_2O_3 + $CaCO_3$ + coke powder) with the help of a crane from the Throat of the furnace.

Combustion Zone :
First introduce carbon from the Throat, carbon comes down, and contacts with hot Air-blown, from down and combustion of carbon stars and carbon dioxide will form. As temperature increases, carbon dioxide convert into carbon mono oxide

Reduction Zone :
Now, carbon monoxide moves upward due to the gaseous state. In the reduction zone Fe_2O_3 reacts with carbon monoxide; by reducing Fe_2O_3 first converts into FeO and then it will reduce in Fe solid because Its temperature is below at meltin Slagint and carbon converts into carbon dioxide.

Slage Formation :
Calcium carbonate convert into Calcium oxide, and it will react with SiO_2 and form

Fusion Zone :
Fe solid now starts melting by moving down due to increases in temperature (1503°C). This melted. Fe is called as **pig iron.**

Q45 **In the metallurgy of iron, when limestone is added to the blast furnace, the calcium iron ends up in-**
(A) Slag (B) Gangue (C) Metallic (D) $CaCO_3$

Sol. **(A)**

Q46 **A flux is often added to remove impurities from an ore in a blast furnace. In the reaction, $CaO + SiO_2 \longrightarrow CaSiO_3$ the Slag and the flux are-**

(A) $CaSiO_3$ and SiO_2 **(B) $CaSiO_3$ and CaO**

(C) CaO and SiO_2 **(D) SiO_2 and $CaSiO_3$**

Sol. **(B)**

Q47 The atmosphere is maintained in the hearth of the blast furnace.

(A) Reducing (B) Oxidising

(C) Neutral (D) Acidic

Sol. (A)

Q48 Exothermic reaction taking place in-

(A) Hearth (B) Combustion zone

(C) Fusion zone (D) Reduction zone

Sol. (B)

Q49 Near the top of a blast furnace employed for the extraction of iron, the metal oxide are reduced to sponge iron by-

(A) Carbon (B) CO

(C) CO_2 (D) Limestone

Sol. (B)

Q50 Iron obtained from blast furnace is-

(A) wrought iron (B) Cast iron

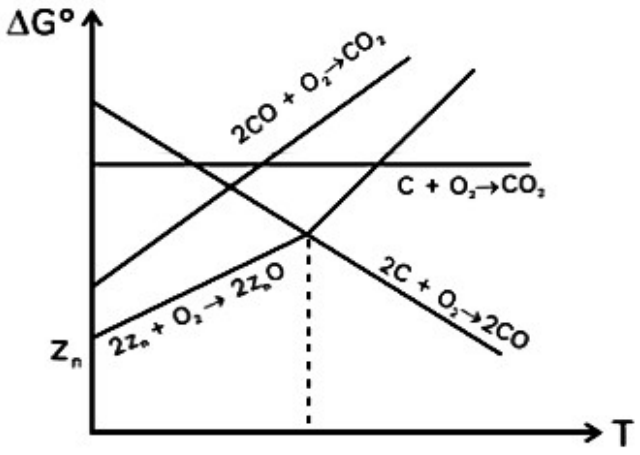
(C) Pig iron (D) steel

Sol. (C)

2. Reduction of Zinc :

Ore of Zn ⇒ ZnS (zinc blend or sphalerite)

Ellingham's diagram shows that at boiling temperature, ZnO is reduced by carbon.



Reduction of zinc takes place in Vertical Retort.

Flow chart

Roasting of zinc blend(ZnS) and we get ZnO

$ZnS \longrightarrow SO_2 + ZnO$

⇓

Reduction

$ZnO + C \longrightarrow CO + Zn(g)$

⇓ (Vapours)

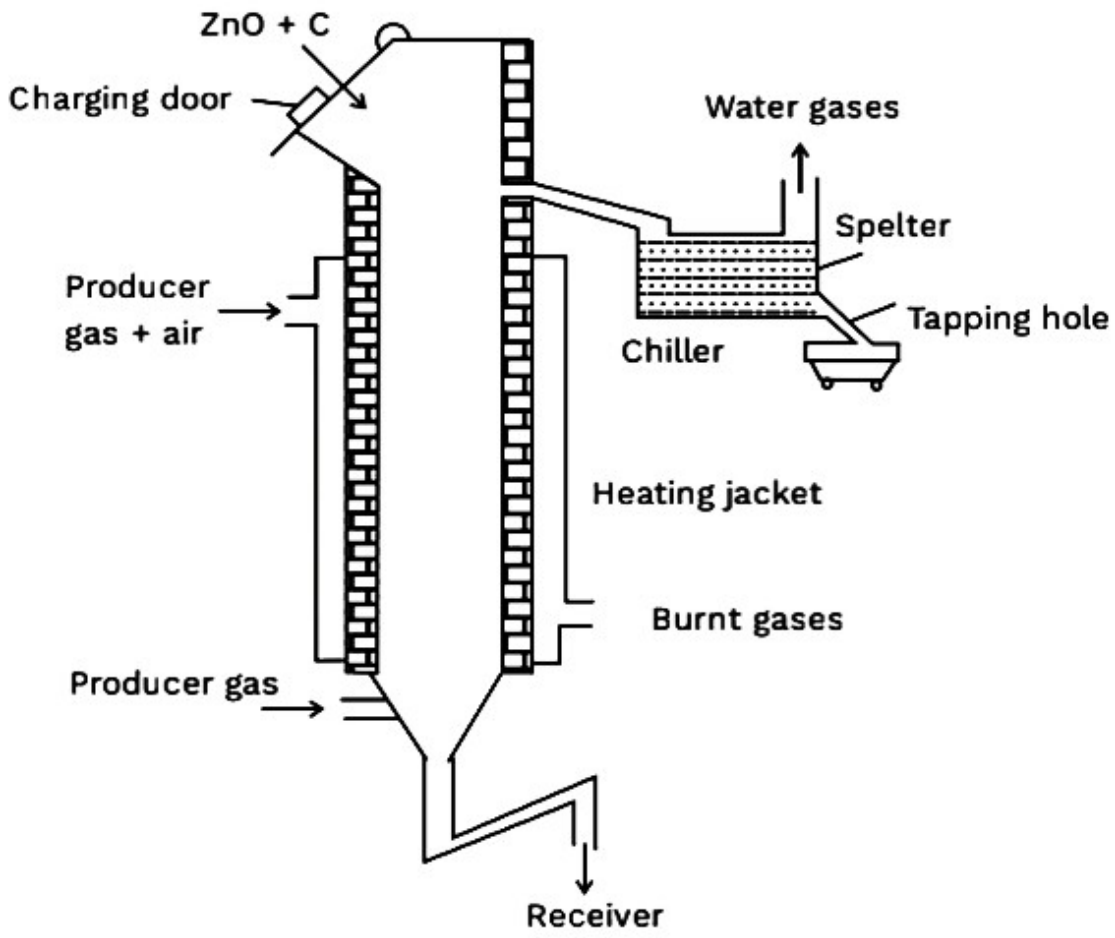
$Zn(g) \Rightarrow$ rapid Cooling

⇓

$Zn(s)$(impure)

It is knowns as **spelter Zinc** (90% pure)

Vertical Retort



Vertical Retort

Working :
Vertical Retort is charged by raw material (ZnO + coke powder). ZnO heated with carbon and produce Zn-vapour

Chiller :
Used for rapid cooling of Zn-vapour is separate out in the form of Zn-liquid and carbon mono oxide removes as fumes.

Heating Jacket :
Fuel contain in this zone for Reduction and heating

Receiver :
Non-Volatile impurities settle down from the receiver.

Q51 **Spelter is:**

(A) impure Cu **(B) impure Zn** **(C) ZnO** **(D) CuO**

Sol. **(B)**

3. Reduction of copper :

Ore of copper ⇒ Chalcopyrite (partially Reduction)

$$\Downarrow$$

$$\text{Copper matte}\left(\underset{98\%}{Cu_2S} + \underset{2\%}{FeS}\right)$$

Q52 **Copper matte consists of**

(A) Copper oxide and ferrous sulphide

(B) Copper sulphide and ferrous oxide

(C) Copper sulphide and ferrous sulphide

(D) Copper oxide and ferrous oxide

Sol. **(C)**

Process :

Reduction of copper ore process takes place in Bessemer converter.

Flow chart :

In the Bessemer converter controlled Air on copper matte (Cu_2S + FeS)

$$\Downarrow$$

Controlled Air passed

Bessemerisation

$$FeS + \underset{air}{O_2} \longrightarrow FeO + SO_2 \uparrow$$

$$\Downarrow$$

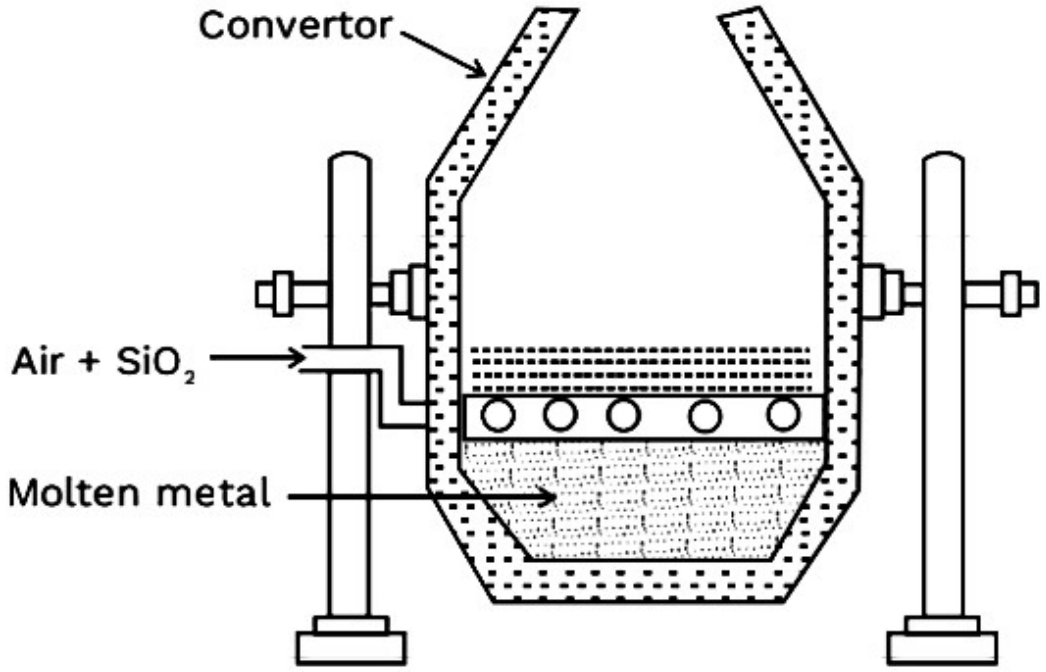
$$FeO + \underset{flux}{SiO_2} \longrightarrow FeSiO_3$$

$$\Downarrow$$

$$Cu_2S + \underset{Limited}{O_2} \longrightarrow CuO_2 + SO_2 \uparrow (slag)$$

$$Cu_2S + Cu_2O \longrightarrow \underset{\substack{\Downarrow \\ \text{Blister Copper} \\ \text{(98\% Pure)}}}{Cu} + SO_2 \uparrow$$

Bessemer Convertor



Bessemer convertor of copper

Working :

In the Bessemer convertor, first Controlled Air passed on copper matte and FeS react and convert into FeO and SO_2

Slage formation :

In the container, we add SiO_2 (Flux) to convert FeO into Slag $FeSiO_3$, which remove from the rotatory Bessemer convertor.

Bessemrisation :

Now bypassing limited Air remaining Cu_2S convert Cu_2O and SO_2 will remove.

Self-reduction :

Cu_2O self-reduced by Cu_2S and form Cu, and it is known as **blister copper** due to the removal of SO_2

Q53 Formation of metallic copper from the sulphide are the commercial thermo-metallurgical Process essentially involves which one of the following reaction:

(A) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2; CuO + C \longrightarrow Cu + CO$

(B) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2; 2Cu_2 + Cu_2S \longrightarrow 6Cu + SO_2$

(C) $Cu_2S + 2O_2 \longrightarrow CuSO_4; CuSO_4 + Cu_2S \longrightarrow 3Cu + 2SO_2$

(D) $Cu_2S + \frac{3}{2}O_2 \longrightarrow Cu_2O + SO_2; Cu_2O + CO \longrightarrow 2Cu + CO_2$

Sol. (B)

Q54 Self-reduction of Cu_2S to Cu can be Carried out in :

(A) Bessemer convertor
(B) blast furnace
(C) both (A) and (B)
(D) None of these

Sol. (A)

Q55 Blister copper is:

(A) impure copper
(B) obtained in self reduction process in bessemerisation
(C) both (A) and (B) are correct
(D) none of above

Sol. (C)

Q56 Cupric oxides is reduced to copper by

(A) H_2 (B) C (C) CO (D) All of these

Sol. (D)

Q57 **In the Smelting of Cu_2S, the flux is-**

(A) $CaSiO_3$ (B) FeO (C) $FeSiO_3$ (D) SiO_2

Sol. (D) $FeS \longrightarrow FeO \xrightarrow{SiO_2} FeSiO_3$

Q58 **In the extraction of copper from its sulphide ore, the metal is formed by the Reduction of Cu_2O with:**

(A) FeS (B) CO (C) Cu_2S (D) SO_2

Sol. (C) Self reduction $Cu_2O + Cu_2S$

Q59 **Blister Cu is about:**

(A) 60% Cu (B) 90% (C) 98% (D) 100% Cu

Sol. (C) Fact

Q60 **Silica is added to roasted copper ores during extraction in order to remove.**

(A) Cuprous sulphide (B) Ferrous oxide

(C) Ferrous sulphide (D) Cuprous oxide

Sol. (B) $FeO + SiO_2 \longrightarrow FeSiO_3$

Q61 **Product obtained after Bessemerisation is called as.....because.**

(A) Concentrated copper; copper percentage is high

(B) Copper matte; of its appearance

(C) blister copper; of its appearance

(D) Ultra pure copper; 100 percent copper

Sol. (C) Fact

Q62 **Which reaction take place in Bessemer converter-**

(A) $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$ **(B)** $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

(C) $CuO + C \longrightarrow Cu + CO$ **(D)** $CuCO_3 \longrightarrow CuO + CO_2$

Sol. (A) The best option is (A) due to Accentually this reaction takes place.

Q63 **In the extraction of the copper, metal is formed in the Bessemer converter due to reaction.**

(A) $Cu_2S + 2CuO \longrightarrow 6Cu + SO_2$ **(B)** $Cu_2S \longrightarrow 2Cu + S$

(C) $Fe + Cu_2O \longrightarrow 2Cu + FeO$ **(D)** $2Cu_2O \longrightarrow 4Cu + O_2$

Sol. (A) Extraction from Cu_2S

Q64 **Slag is obtained during the smelting process in the extraction of copper from copper pyrite is composed mainly of:**

(A) Cu_2S **(B)** $FeSiO_3$ **(C)** $CuSiO_3$ **(D)** SiO_2

Sol. (B) $FeO + SiO_2 \longrightarrow FeSiO_3$

4. Reduction of lead

Ore of lead $\Rightarrow$ PbS (Galena)

$$PbS + O_2 \xrightarrow{T_1} PbO \downarrow \text{ (Partial roosting)}$$

$$[\underset{\text{(major)}}{PbO} \text{ or } \underset{\text{(Small)}}{PbSO_4} \text{ (rarely)}]$$

Process :

After Partial roasting of PbS we get PbO which again reacts with PbS, and By self-reduction, we will get Pb.

Self-reduction $\begin{cases} PbO + PbS \xrightarrow{T_1} \underset{\text{(impure)}}{Pb} + SO_2 \\ PbSO_4 + PbS \xrightarrow{T_2} Pb + SO_2 \end{cases}$ (Always $T_2 > T_1$)

Q65 **Identity the metal M whose extraction is based on the following reaction:**

$MS + 2O_2 \longrightarrow MSO_4$; $2MS + 3O_2 \longrightarrow 2MO + 2SO_2$

$MS + 2MO \longrightarrow 3M + SO_2$; $MS + MSO_4 \longrightarrow 2M + 2SO_2$

(A) Magnesium **(B) Aluminium** **(C) Lead** **(D) Tin**

Sol. (C)

Q66 **Which of the following reactions represents the self-reduction Process?**

(A) $\begin{cases} HgS + O_2 \longrightarrow HgO + SO_2 \\ HgO + HgS \longrightarrow Hg + SO_2 \end{cases}$ **(B)** $\begin{matrix} Cu_2S + O_2 \longrightarrow Cu_2O + SO_2 \\ Cu_2S + Cu_2O \longrightarrow Cu + SO_2 \end{matrix}$

(C) $\begin{matrix} PbS + O_2 \longrightarrow PbO + SO_2 \\ PbO + PbS \longrightarrow Pb + SO_2 \end{matrix}$ **(D) All of these**

Sol. (D)

Q67 **Main Source of lead is PbS, It is converted to Pb by:**

(X) : $PbS \xrightarrow[\Delta]{air} PbO + SO_2$

$\xrightarrow{C} Pb + CO_2$

(Y) : $PbS \xrightarrow[\Delta]{air} PbO + PbS$

$\longrightarrow Pb + SO_2$

(Z) : $PbS \xrightarrow[\Delta]{air} PbO + SO_2$

$\xrightarrow{CO} Pb + CO_2$

Self-Reduction Process is:

(A) (X) **(B) (Y)** **(C) (Z)** **(D) None**

Sol. (B)

Q68 **In which of the following solutions no reducing agent is required:**

(A) iron from hematite **(B) Tin from cassiterite**

(C) mercury from cinnabar **(D) Tin from zinc**

Sol. **(C)**

Note

This is the additional part related to JEE advance.

For Low-grade (less content of lead) core of lead, carbon reduction is used.

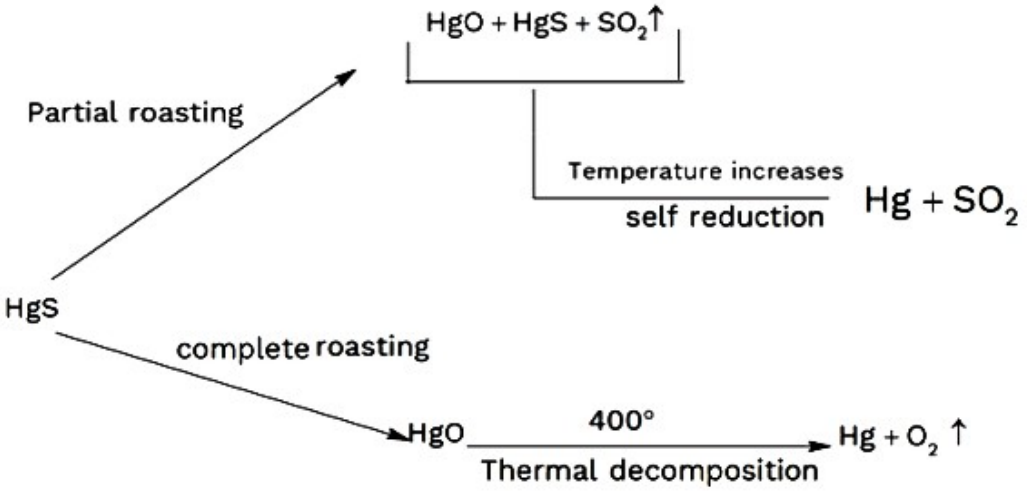
$$\underset{\text{(Low grade)}}{PbS} + O_2 \xrightarrow{\text{Complete}} PbO + SO_2 \xrightarrow{C} Pb + CO_2$$

Reason ⇒ Due to impurities are High, so that Controlled roosting is not possible. So self-reduction is not possible.

5. Reduction of mercury (Hg):-

Ore of mercury (Hg) ⇒ HgS

Process : We can do both partial roasting as well as complete roasting for HgS.



Working :
In partial roasting, give Hg and SO_2 due to self-reduction at high temperatures.
In complete roasting, HgO will form, and by taking 400°C temperature (decomposition temperature of HgO), it gives Hg and O_2. This Process is named as thermal decomposition, and there is no need for any reducing agent.

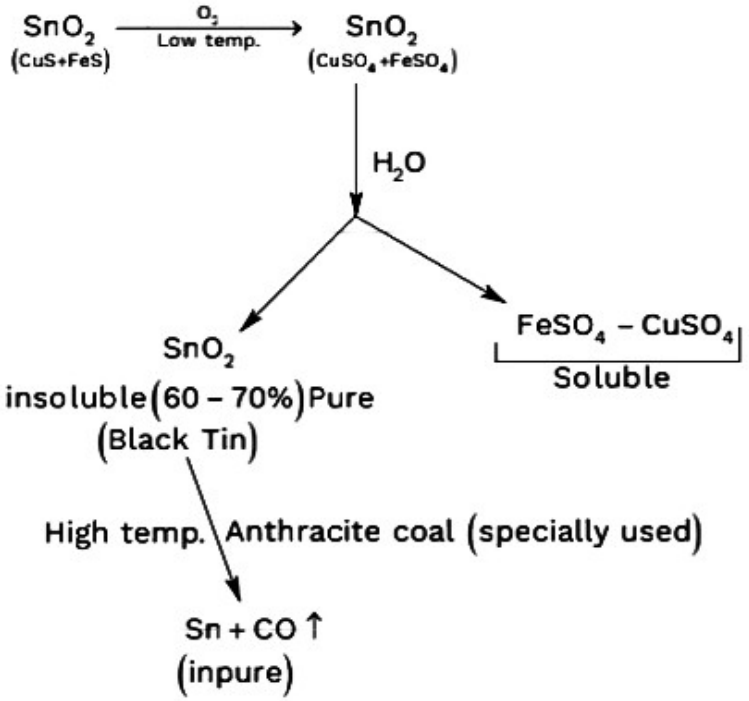
6 Reduction of Tin:

Ore of Tin (Sn) ⇒ Black Tin (SnO_2)

Impurity ⇒ CuS+FeS

Process:
Roasting is carried out due to impurities present as CuS and FeS.

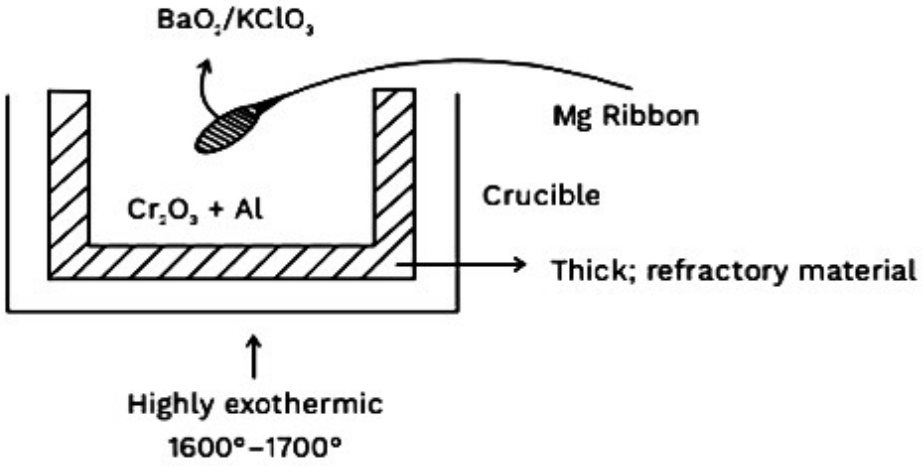
Working:
By roasting CuS and FeS convert into sulphates at low temperatures. Then, by adding water sulphate of Cu and Fe separate as soluble form and SnO_2 as insoluble form and it is 60-70% pure and known as **Black Tin**.
To Reduction of this impure Black Tin, we use Anthracite coal and by heating carbon mono oxide and Pure Tin will form.



7. Reduction of Cr (Chromium) and Mn [GoldSchmidt Alumino thermite process]

(I) Reduction of Cr : Ore of Cr $\Rightarrow$ Cr_2O_3

Process: We will take crucible, which is absorbed high temperature & made by bone ash.



$$KO_2 / BaO_2 \xrightarrow{\Delta} O_2 \xrightarrow{Mg} MgO + Energy$$

$$Cr_2O_3 + Al \longrightarrow Al_2O_3 + Cr + Energy$$

Working : In the crucible to initiate the reaction, we will take the help of an ignition mixture of Mg + BaO_2 /KO_2. They decompose and release oxygen, which reacts with Mg and form MgO, and a huge amount of energy will be used in the reaction of Cr_2O_3 and Al and Cr will reduce and Al_2O_3 form along with huge energy.

(II) Reduction of G $_2$ (Thermic process)

Ore of Mn $\Rightarrow$ MnO_2

$$MnO_2 + Al \longrightarrow Al_2O_3 + Mn\,(\text{explosive reaction})$$
$$(\text{due to excess of oxygen})$$

$$MnO_2 \xrightarrow[-O_2]{\Delta} Mn_3O_4 \xrightarrow{Al} Al_2O_3 + Mn$$

Working: MnO_2 reaction with Al is an explosive reaction due excess of Oxygen, so first MnO_2

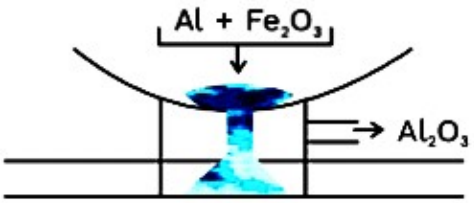
is heated and convert Mn_3O_4 and then reacts with Al and give Mn. This Process is also called as Thermite process, which use thermite welding.

Thermite welding :

This Process is used for welding for railway tracks. Thermite mixture will take in 1 : 3 of $Al : Fe_2O_3$

$Al : Fe_2O_3$ (Thermite mixture)

1 : 3 (by mass)



Railway Track
(Thermite Welding)

Q69 **In the alumino thermite process, Al Act as**
(A) An oxidizing agent (B) A flux
(C) A reducing agent (D) A solder

Sol. **(C)**

Q70 **Assertion: Al is used as a reducing agent in Aluminothermy:**
Reason: Al has a lower melting point than Fe, Cr, and Mn.
(A) S-1 is true, S-2 true and S-2 is the correct explanation of S-1
(B) S-1 is true, S-2 is true, and S-2 is not the correct explanation for S-1
(C) S-1 is true, S-2 is true
(D) S-1 is false, S-2 is true

Sol. (B)

Q71 **Thermite is a mixture used for welding is:**

(A) Fe and Al
(B) Ferric oxide and Aluminium power
(C) Barium Pergolide and Magnesium power
(D) Cu and Aluminium

Sol. (B)

Q72 **In the alumina thermic process aluminium act as:**

(A) An oxidising agent **(B) A flux**
(C) A reducing agent **(D) Solder**

Sol. **(C)**

Q73 **For the extraction of chromium from Cr_2O_3, the Process adopted is-**

(A) Carbon Reduction **(B) Alumino thermite**
(C) Electrolytic Process **(D) None**

Sol. **(B)**

Q74 **Reducing metal required in Gold Schmidt Alumino thermic Process is-**

(A) Chromium powder **(B) Zinc powder**
(C) Aluminium powder **(D) Silver powder**

Sol. **(C)**

Q75 **The Substance used in the Thermite Process for reducing metal oxide.**

(A) Aluminium powder **(B) Thorium**
(C) Heated ft **(D) Carbon**

Sol. **(A)**

8. Reduction of Aluminium [Electrolytic reduction]

Highly reactive metals are reduced by electrolysis by their molten salts as we learn in metal reactivity series. Aluminium is highly reactive metal so it is reduced by electrolysis.
For s-block or p-block element electrolysis takes place with chloride generally, Bromide & iodied in used due to covalent in nature. In case of fluoride, gas release & its highly corrosive in nature.

Process :
In case of Al (aluminium) and Be (berelium) oxides are used because chlorides ($AlCl_3$ and $BeCl_2$) has covalent character. Electrolytic reduction takes place in fused state. It is done by Hall-Heroult's process.

Oxide of Aluminium – Al_2O_3 (M.P. 2030°C)

Solvent used – Na_3AlF_6 Cryolite (M.P. 1013°C)

Electrolytic mixture:	**Na_3AlF_6**	**: Al_2O_3**	**: CaF_2**
	85%	**10%**	**5%**

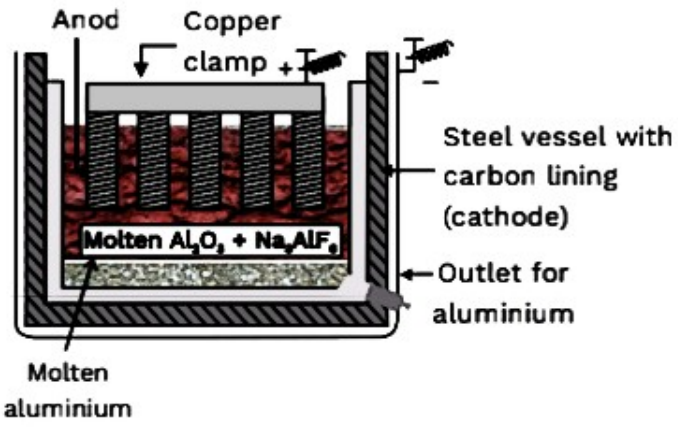
Solvent is used in this process due to high melting point of Al_2O_3, solvent reduce working temperature and increase conductivity.

Note

Sometimes CaF_2 (fluorspar) is also used with electrolytic solution because it reduces working temperature up to 950°C and increase conductivity.

Working :
Electrolysis takes place by Hall-Heroult's process which is a electrolytic cell container.
In this process the density of Aluminium is greater (density ⇒ Al_2O_3 + Na_3AlF_6 < Al) so that, Al reduced at cathode and settled and oxygen gas release at anode.

Reaction involve:
Cathode ⇒ $3e^- + Al^{+3} \rightarrow Al\downarrow$
Anode ⇒ $O^{-2} \rightarrow O_2 \uparrow + 2e^-$



Electrolytic cell for the extraction of aluminium

O_2 reacts with anode from CO/CO_2, so to protect anode from oxygen, we used powder of carbon. This process as deterioration of anode.
The addition of C-powder is also used to decreases heat release.
Solubility of Al_2O_3 in Na_3AlF_6 is 15% of Na_3AlF_6

Function of CaF_2 in electrolytic mixture :

$$\left|\begin{matrix} Al_2O_3 + CaF_2 \\ Na_3AlF_6 \end{matrix}\right| \quad \begin{matrix} \text{M.P. } CaF_2 < Al_2O_3 \\ \text{ionisation } Ca^{+2}F^{\ominus} > Al^{+3}O^{-2} \end{matrix}$$

Anode $\Rightarrow$ $2F^- \rightarrow F_2 + 2e^-$

$$F_2 + O^{-2} \rightarrow O_2\uparrow + F\ominus \xrightarrow[C]{\text{Anode}} Co / CO_2 \uparrow$$

Cathode $\Rightarrow$ $3e^- + Al^{+3} \rightarrow Al$

Q76 **A metal has a high concentration into the earth crust and whose oxides cannot be reduced by carbon. The most suitable method for the extraction of such metal is-**

(A) Alumino thermite process
(B) Electrolysis process
(C) Van-Arkel's process
(D) Cupellation.

Sol. **(B)**

Q77 The element which could be extracted by electrolysis reduction of its oxide dissolved in a high temperature melt is

(A) Sodium (B) Magnesium (C) Flourine (D) Aluminium

Sol. (D)

Q78 Which one of the following metals is extracted by carbon reduction process-

(A) Sodium (B) Iron (C) Aluminium (D) Magnesium

Sol. (B)

Q79 Electrolytic reduction method is used for the extraction of-

(A) Highly electronegative elements (B) Highly electro positive element

(C) Transition metal (D) Metalloids

Sol. (B)

Q80 Aluminium is obtained by

(A) Reducing Al_2O_3 with coke

(B) Electrolysing Al_2O_3 dissolved in Na_3AlF_6

(C) Reducing Al_2O_3 with cryolite

(D) Heating Al_2O_3 and cryolite

Sol. (B)

Q81 Electrolytic reduction process is used for the altercation of:

(A) Alkali metal (B) Alkaline earth metal

(C) Aluminium (D) All of the above

Sol. (D)

Q82 In the electrolysis of Alumina, Cryolite is added to:
(A) Lower the melting point of aluminium
(B) increases the electrical conductivity
(C) Both (A) and (B)
(D) Highly electropositive elements.

Sol. (C)

Q83 The electrolytic reduction technique is used in the extraction of:
(A) High electronegative element (B) Transition metal
(C) Metalloids (D) Highly electropositive elements

Sol. (D)

Q84 Which of the following mixture is used in the electrolysis of Aluminium -
(A) $Al_2O_3 + 8aF_2$ (B) $Na_3AlF_6 + Al_2O_3$
(C) $NaAlO_2 + Al_2O_3$ (D) $Al_2O_3 + NaOH$

Sol. (B)

Q85 In the Hall-Heroult process for extraction of Al, the Ore is fused with-
(A) $NaHCO_3$ (B) Na_2CO_3
(C) NaF (D) Na_3AlF_6

Sol. (D)

Q86 In the commercial electrochemical process for aluminium extraction the electrolyte used for:

(A) $Al(OH)_3$ in NaOH solution

(B) An aqueous solution of $Al_2(SO_4)_3$

(C) A molten mixture of Al_2O_3, Na_3AlF_6 & CaF_2

(D) A Molten mixture of Al_2O_3 and $Al(OH)_3$

Sol. (C)

Q87 During electrolytic reduction of Alumina, two auxilimary electrolytes X and Y are added to increase the electrical conductance and lower the temperature of melt in order to making fused mixture very conducting.

(A) Cryolite and fluorspar (B) Cryolite and alum

(C) alum and fluorspar (D) fluorspar and bauxite

Sol. (B)

Q88 In electrolysis of Al_2O_3 by Hall-Heroult process.

(A) Cryolite $Na_3[AlF_6]$ lowers the melting point of Al_2O_3 and increasing its electrical conductivity

(B) Al is obtained at cathode and probably CO_2 at anode

(C) both (A) and (B) are correct

(D) None

Sol. (C)

Q89 During the electrolytic reduction of aluminium, the carbon anodes are replaced from time to time because

(A) The carbon anodes get decayed

(B) The carbon prevents atmospheric oxygen from coming in contact with aluminium

(C) oxygen liberated at the carbon anodes reacts with anode to form CO and CO_2

(D) Carbon converts Al_2O_3 to Al

Sol. (C)

Refining/Purification

Reduction of ore give metal by removing some non-volatile impurities but some of the impurities are remaining so that
the final step is refining/purification of metal. There are various methods for refining of metals.

Distillation:

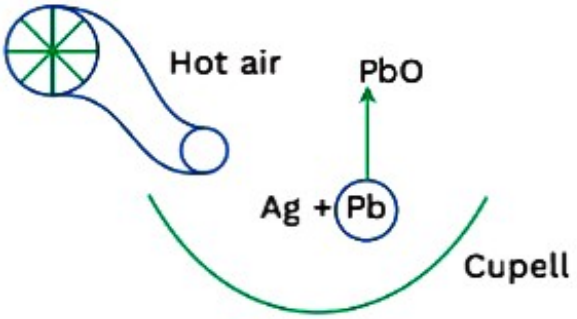
Process ⇒ Heating of metal just above its boiling point so impurities remain in solidly state
Used for ⇒ Zn, Cd, Hg
Concept ⇒ When boiling point of pure metal less than B.P. of impurities.
Function ⇒ Zn, Cd, Hg used due to $d^{10}s^2$ full filled orbital, d-d metal-metal bonding not possible so weak attraction is present So boiling point decreases.

2. Liquation:

Process ⇒ impure metal heated on sloped surface furnace, pure metaobtain in liquid state, impurities remains in solid state.
Used for ⇒ Sn, Pb, Bi
Concept ⇒ Melting point Pure metal is less than of impurities.
Function ⇒ Sn, Pb & Bi heavy p-block metal in this metal-metal bonding decrease, size increase and metallic character decreases.

3. Cupellation :

When silver contain impurities of Pb
Process ⇒ it is an oxidative refining.
Used for ⇒ Ag
Concept ⇒ In a cupell, hot air passed and silver do not react with oxygen at high temperature but Pb react with O_2 form volatile PbO
Function ⇒ Silver impurities do not react because less reactivity of Ag.



$$Ag \xrightarrow{\Delta + O_2} X(\text{no reaction})$$

$$Pb \xrightarrow[\text{Hot air}]{\Delta + O_2} pbO \uparrow (\text{valatile})$$

4. Zone Refining :

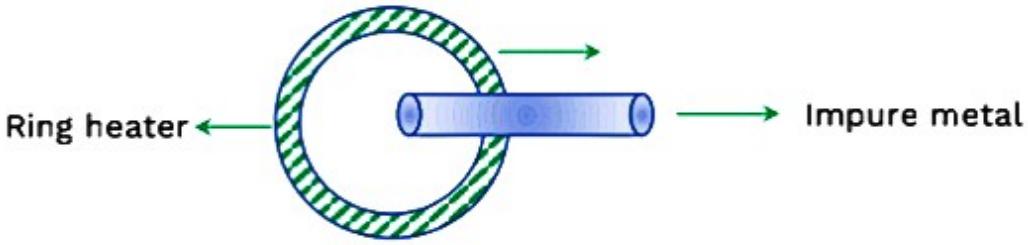
Process ⇒ To obtain metal/semimetals in ultra-pure form.

Used for ⇒ Semiconductors like Si, B, Ga, Ge, As, In

Concept ⇒ This process is based on Phase rule

⇓

Impurities are more soluble in molten state in comparison of solid-state crystallization



Function ⇒ As ring heater move molten impurities move along heater and remove in the last and remains pure metal.

5. Vapour phase refining

Process ⇒ Metal form volatile compound in presence of volatile reagent and This process based on temperature difference.

Used for ⇒ metals like Ni, Zr, Ti

Function ⇒ impure metal + Reagent (volatile) are react with each other and metal & reagent vaporised at T_1 temperature and impurities remains. Then-Metal and Reagent heated at T_2 temperature, so reagent vaporised and pure metal remains.

Example:

(a) Mond's Process: purification of nickel

$$Ni_{(s)} + CO_{(g)} \xrightarrow{40°-50°} \underset{\text{(impure)}}{Ni(CO_2)_{(2)}} + \text{impurities}$$

$$Ni(CO)_{4(s)} \xrightarrow{200°-220°C} \underset{\text{Pure}}{Ni(s)} + CO_{(g)}\uparrow$$

(b) Van-Arkel's Process:

Process ⇒ Van-Arkel's process is used electrical heating with iodine.

Use for ⇒ Ti, Zn, Hf, B

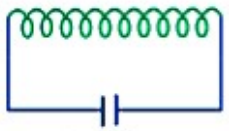
Function ⇒ impure metal heated with iodine at 400K temperature and Mix form which heated with tungsten filament at high temperature approx. 1400K and

pure metal extracted.

$$M + I_2 \xrightarrow{\approx 400K} MI_x + \text{impurities}$$

(impure)

$$MI_x \xrightarrow{1400K} M + I_2\uparrow$$



(Pure)

Coiling

6. Park's Process:

Process ⇒ De-silverization of Argentiferous Lead

Used for ⇒ Pb – Ag (content of Pb >>> Ag)

Concept ⇒ Based on alloy formation tendency of d-block metals and distribution low (that is solubility of a metal in different solvent) or partition coefficient (ration of solubility).

Function ⇒ Ag is 3000 times more soluble in zinc comparison of lead.

Partition coefficient for Ag in Zn/Pd = 3000

⇒ Zn ⇒ 4th period → d-block

Ag ⇒ 5th period → d-block

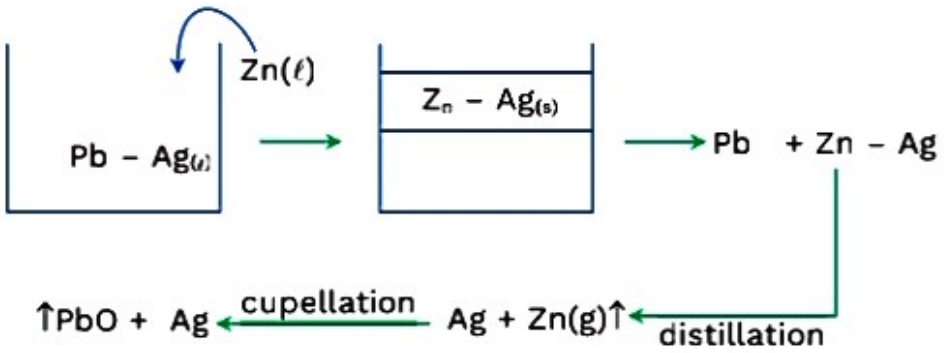
Pb ⇒ 6th period → p-block

We know that almost same size elements easily form alloy

means Zn – Ag ⇒ Alloy form

Ag – Pb ⇒ Alloy form

Zn – Pb ⇒ no alloy formation



When We add Ag – Pb liquid. With Zn(ℓ), then due to density Pb separated and Zn – Ag give Ag by distillation but the Ag have some impurity of Pb so now cupellation takes place and get pure Ag.

7. Pattison's Process.

Process ⇒ De-silverization process of Argentiferous lead.

Used for ⇒ Pb – Ag (content of Pb >>> Ag)

Concept ⇒ Based on fractional crystallization

M.P. of Pb – Ag alloy = 327°C

M.P. of Pure Pb = 302°C



$$Pb - Ag_{(s)} \xrightarrow{>327^{\circ}} Pb - Ag_{(\ell)} \xrightarrow{327^{\circ} >> T >> 302} Pb_{(s)} + (Ag - Pb) \xrightarrow{\text{Cupellation}} Ag_{(Pure)}$$

Function ⇒ Impure Pb is Heated above M.P. of pure Pb then temperature is maintained between 327° and 302°C so Pb convert into solid form and after several repetition in liquid, Ag present as impurity, silver obtained by cupellation.

8. Amalgamation:

Process ⇒ Hg alloy is called amalgam (As like gravity separation)

Concept ⇒ This process based on difference in density used for Ag and Au (heavy metals)

Note

Pt and Fe do not form Amalgam

Function ⇒ Steam of Hg is used, so Ag and Au settle down & impurities comes up-word and by distillation Ag and Au separated and Hg vaporized.

9. Electro refining:

Process ⇒ This is electrolytic process in molten and aqueous condition.

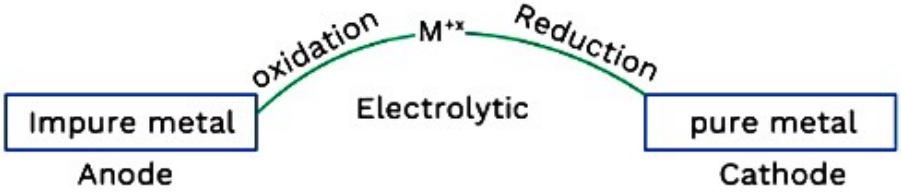
Used for ⇒

Molten state	**Aqueous medium**
For s-block + Al	except s-block/Al Fe, Hg ↓ ↓ Not used (distillation) in pure form

Concept ⇒ Discharge potential (competition between ions to discharge)

Cations $A_u^{+3} < Fg^{+3} < C_u^{+2} ....< Al^{+3} <$ s-block. (It is based on metareactivity)

Function ⇒ For aqueous medium



Example : Electro refining of Cu

Electrolytic mixture – $CuSo_4$ + H_2SO_4 (impure Cu with Ag, Au, Zn, Fe impurities)

Function ⇒ Due less discharge potential Cu first reduce at cathode and impurities are oxidise. Less electropositive impurities oxidise as Anode mud and more electropositive impurities as soluble as cation.

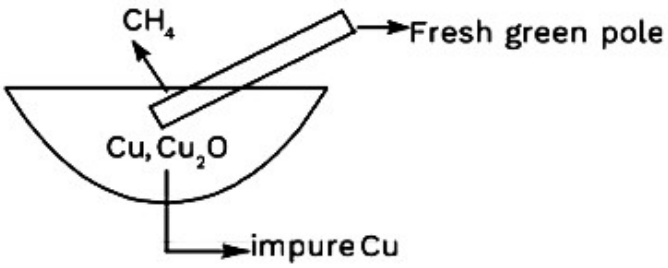
Reaction ⇒ At cathode - $Cu^+ + 1\ e^-$ ® Cu (pure)

At anode – less soluble impurities oxidise as Anode mud.

10. Poling:

Presses ⇒ Refining by Reduction in presence volatile agent (source of fresh green pole which gives unburnt CH_4).

Used for ⇒ Cu, Sn, Pb (they contain impurity of their oxide Cu_2O, SnO_2, PbO)



Function ⇒ The oxide converts in to their respective metal and with fresh green pole use and it give CH_4 which is volatile reducing agent so CO_2 produce which is volatile in nature.

$$Cu_2O \longrightarrow Cu$$

$$PbO \longrightarrow Pb$$

$$SnO_2 \longrightarrow Sn$$

and $CH_4 \longrightarrow Co_2 + H_2O$

Q90 Refectory materials we are generally used in furnaces because :
(A) They are chemically inert
(B) They can withstand high Temperature.
(C) They do not contain impurities
(D) They decrease melting point of ore.

Sol. (B)

Q91 Which of the following statements is correct regarding the slag formation during the extraction of a metal like copper or iron.
(A) The slag is lighter and has lower melting temperature than the metal.
(B) The slag is Havier and has lower melting temperature than the metal.
(C) The slag is lighter and has higher melting temperature than the metal.
(D) The slag is Havier and has higher melting temperature than the metal.

Sol. (A)

Q92 Assertion: Generally, in smelting, roasted/Calcinated Ore is heated with powdered cake in presence of a flux.
Reason: Oxides are reduced to metals by C or Co. impurities are removed as slag.
(A) S-1 is true, S-2 is true and S-2 is correct explanation for S-1
(B) S-1 is true, S-2 is true and S-2 is not correct explanation for S-1.
(C) S-1 is true, S-2 is false
(D) S-1 is false; S-2 is true.

Sol. (A)

Q93 Assertion: Magnesia and quick are used as basic flux.
Reason: MgO and CaO can with stand very high temperature.
(A) S-1 is true, s-2 is true and S-2 is correct explanation for S-1
(B) S-1 is true, S-2 is true and S-2 is not correct explanation for S-1.
(C) S-1 is true, S-2 is false
(D) S-1 is false; S-2 is true.

Sol. (B)

Q94 Assertion: Wolframite impurity is separated from SnO_2 by magnetic separation.
Reason: Tin stone is ferromagnetic, therefore attracted by magnetic.
(A) S-1 is true, s-2 is true and S-2 is correct explanation for S-1
(B) S-1 is true, S-2 is true and S-2 is not correct explanation for S-1.
(C) S-1 is true, S-2 is false
(D) S-1 is false; S-2 is true.

Sol. (C)

Q95 To obtain chromium from chromic oxide $\left(Cr_2O_3\right)$ the method is:

(A) Carbon reduction
(B) Carbon mono oxide reduction
(C) Alumina thermic
(D) Electrolytic process

Sol. (C)

Q96 Selection of temperature to carry out a reduction process depends so as to make.
(A) negative (B) ΔG positive (C) ΔH negative (D) ΔH positive

Sol. (A)

Q97 Ellingham diagram represents.
(A) Change of ΔG with temperature
(B) Change of ΔH with temperature
(C) Change of ΔG with pressure
(D) Change of $\left(\Delta G - T\Delta S\right)$ with temperature.

Sol. (A)

Q98 Which of the following represents the thermite reaction?

(A) $3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$

(B) $MgCO_3 + SiO_2 \longrightarrow MgSiO_3 + CO_2$

(C) $Cu_2S + 2Cu_2O \longrightarrow 6C_4 + SO_2$

(D) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

Sol. (A)

Q99 Cyanide process is used for the extraction of

(A) Au (B) Cu (C) Zn (D) Fe

Sol. (A)

Q100 Extraction of silver from Ag_2S by the use of sodium cyanide is an example of

(A) Roasting (B) Hydrometallurgy

(C) Electrometallurgy (D) Smelting

Sol. (C)

Q101 Which of the following metal can be extracted by smelting

(A) Al (B) Mg (C) Fe (D) None

Sol. (C)

Q102 Electromagnetic separation is used in the concentration of

(A) Copper pyrites (B) Bauxite (C) Cassiterite (D) Cinnabar

Sol. (C)

Q103 Sulphide ore are generally concentrated by

(A) Froth floatation (B) Coasting

(C) Gravity (D) Reduction by carbon

Sol. (A)

Q104 Formation of $Ni(CO)_4$ and subsequent its decomposition into Ni and CO (re-cycled) makes basis of Mond's process.

$$Ni + 4CO \xrightarrow{T_1} Ni(CO)_4 \xrightarrow{T_2} Ni + 4CO$$

(A) 100°C, 50°C (B) 50°C, 100°C (C) 50°C, 230°C (D) 230°C, 50°C

Sol. (C)

Q105 Zone refining is based on the principle of.

(A) Fractional distillation (B) Fractional crystallization

(C) Partition coefficient (D) Chromatographic separation

Sol. (B)

Q106 Si and Ge used for semiconductors are required to be of high purity.

(A) Zone refining

(B) Electro refining

(C) Van-Arkel's process

(D) Cupellation process.

Sol. (A)

Q107 Refining of silver is done by:
(A) Liquation
(B) Poling
(C) Cupellation
(D) Van-Arkell method

Sol. (C)

Q108 Mercury is purified by:
(A) Passing through dilute HNO_3
(B) Distillation
(C) Distribution
(D) Vapour phase refining

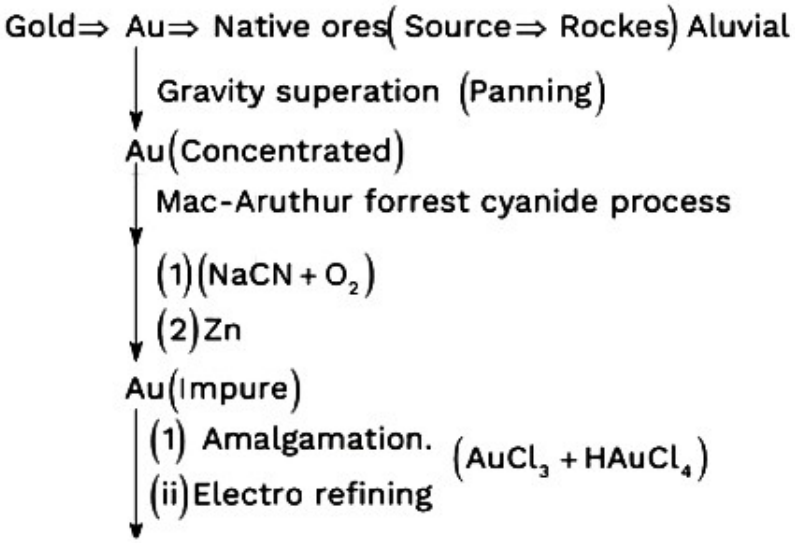
Sol. (B)

Q109 Poling process:
(A) Reduces SnO_2 to Sn
(B) Oxidises impurities like iron removed as scum
(C) Uses green poles
(D) All of above

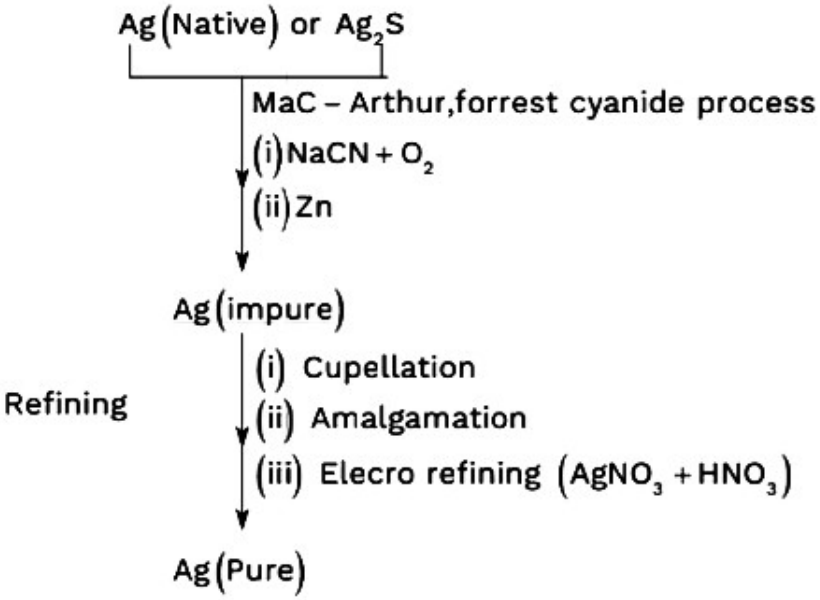
Sol. (D)

Individual metallurgy

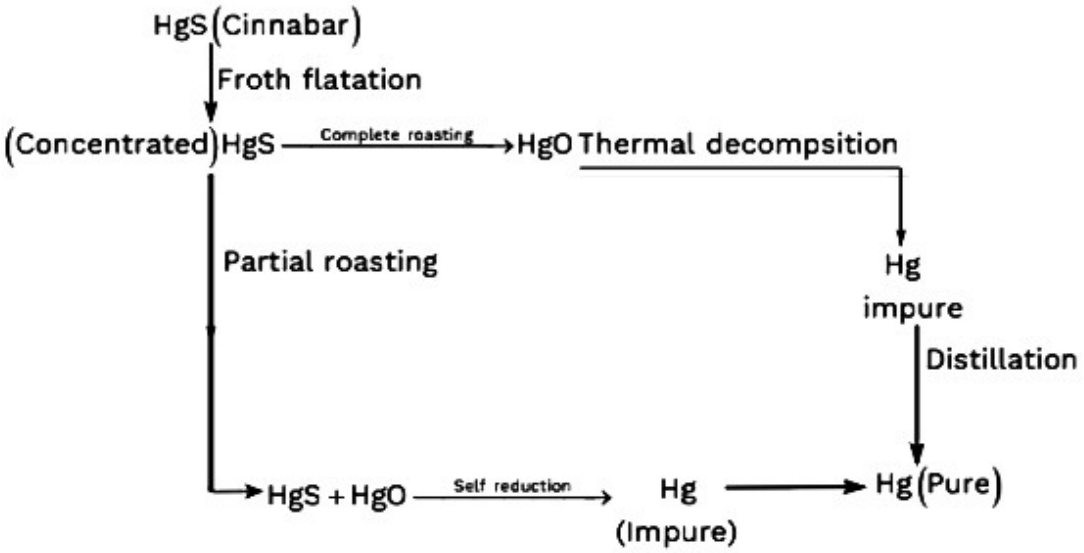
1. For Gold (Au) :



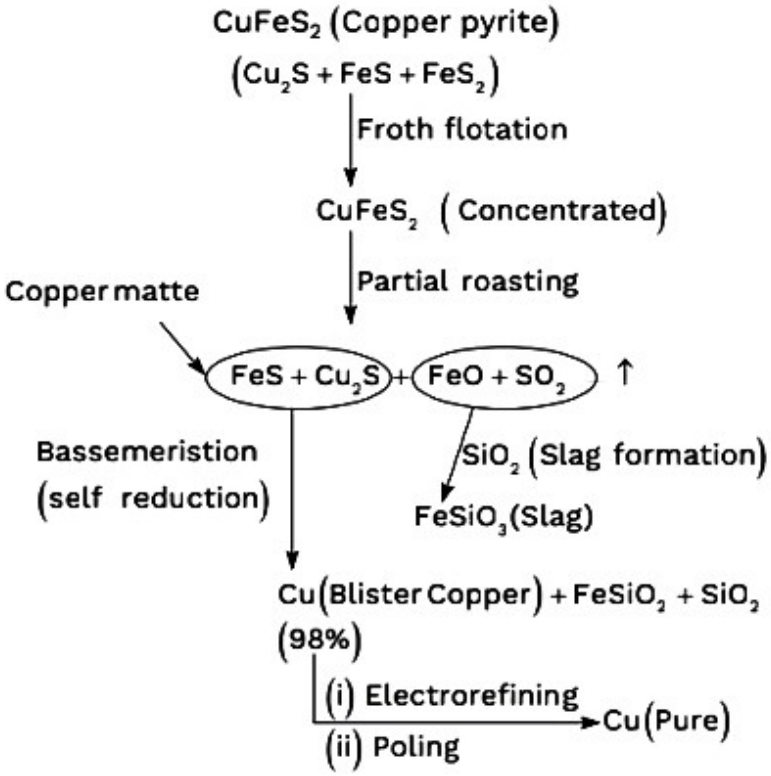
2. Silver :



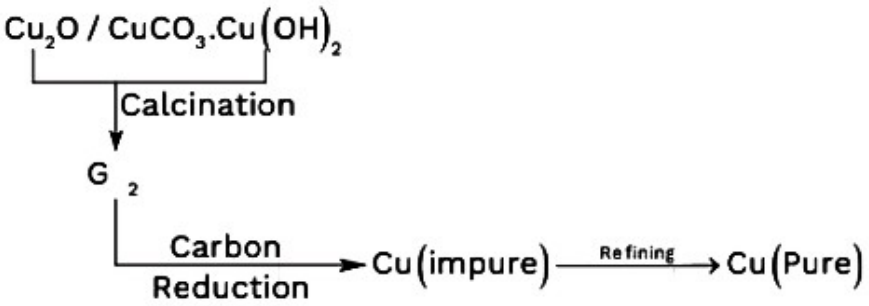
3. Mercury (Hg)



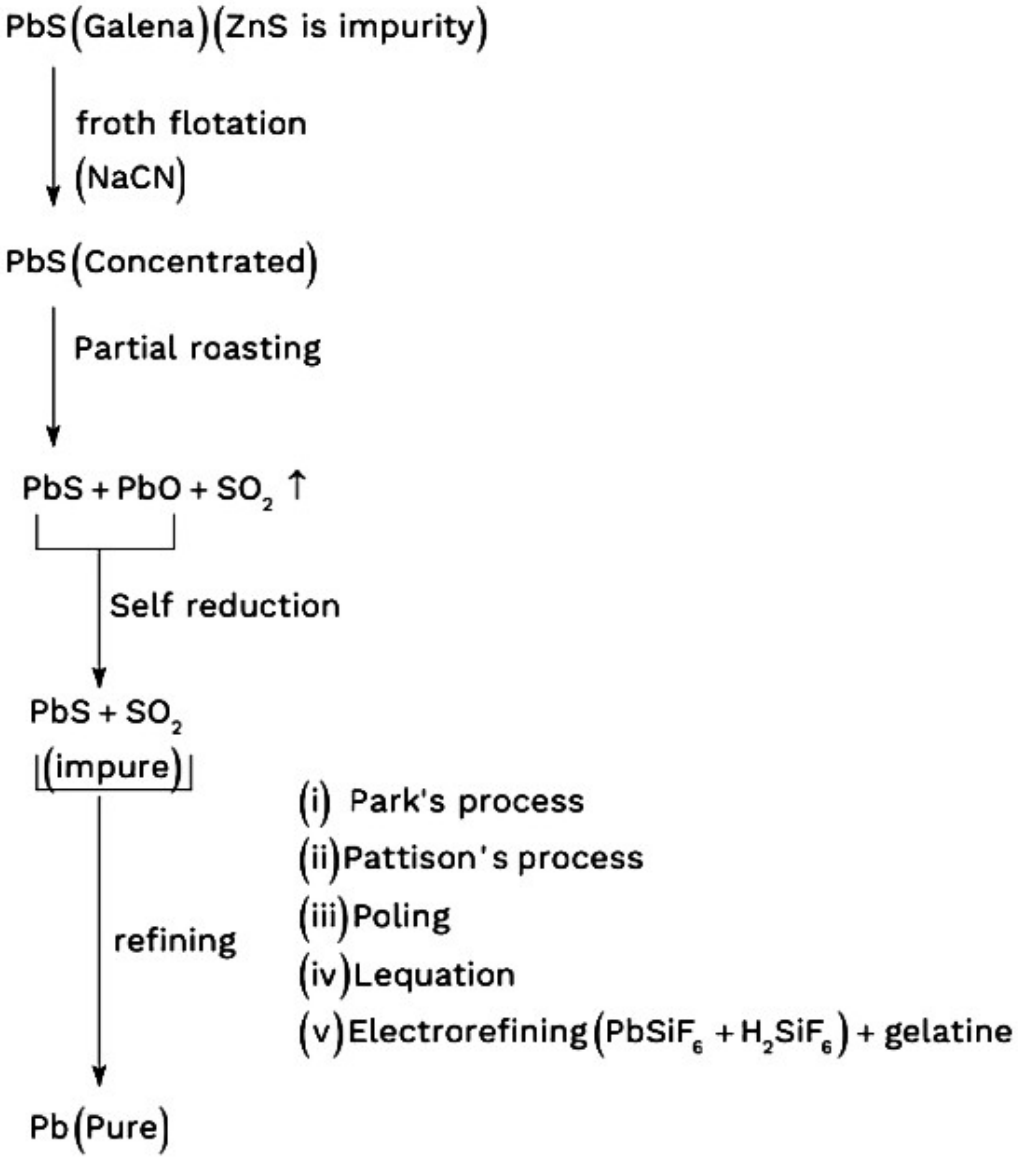
4. Copper (Cu) :



Note : Low grade ore of copper ⇒ Hydrometallurgy


Cu_2O / $CuCO_3.Cu(OH)_2$
Calcination
G 2
Carbon Reduction
Cu (impure)
Refining
Cu (Pure)

5. Lead (Pb) :


PbS (Galena) (ZnS is impurity)
froth flotation
(NaCN)
PbS (Concentrated)
Partial roasting
PbS + PbO + SO_2 ↑
Self reduction
PbS + SO_2
(impure)
refining
(i) Park's process
(ii) Pattison's process
(iii) Poling
(iv) Lequation
(v) Electrorefining ($PbSiF_6$ + H_2SiF_6) + gelatine
Pb (Pure)

6. Tin (Sn) :

SnO_2 (Tin stone)

↓ magnetic seperation

wolframite $(FeWO_4 + M_nWO_4)$ magnetic

SnO_2 (Concentrated)

↓ low temp Roasting → $SnO_2 + C_4SO_4 + FeSO_4 \xrightarrow{H_2O} SnO_2 \downarrow CuSO_4$

$SnO_2 \downarrow + (CuSO_4, FeSO_4)_{aq}$

(Black tin)

↓ Anthracite coal

Sn (impure) ↓ Refining → (i) Poling, (ii) Liquation, (iii) Electroreting

Sn (Pure)

7. Zinc (Zn) :

Zinc blend (ZNS)

↓ froth floatation $(CuSO_4$, as Activator)

ZnS (Concentrated)

↓ roasting

$ZnO + SO_2$

↓ Carbon reduction

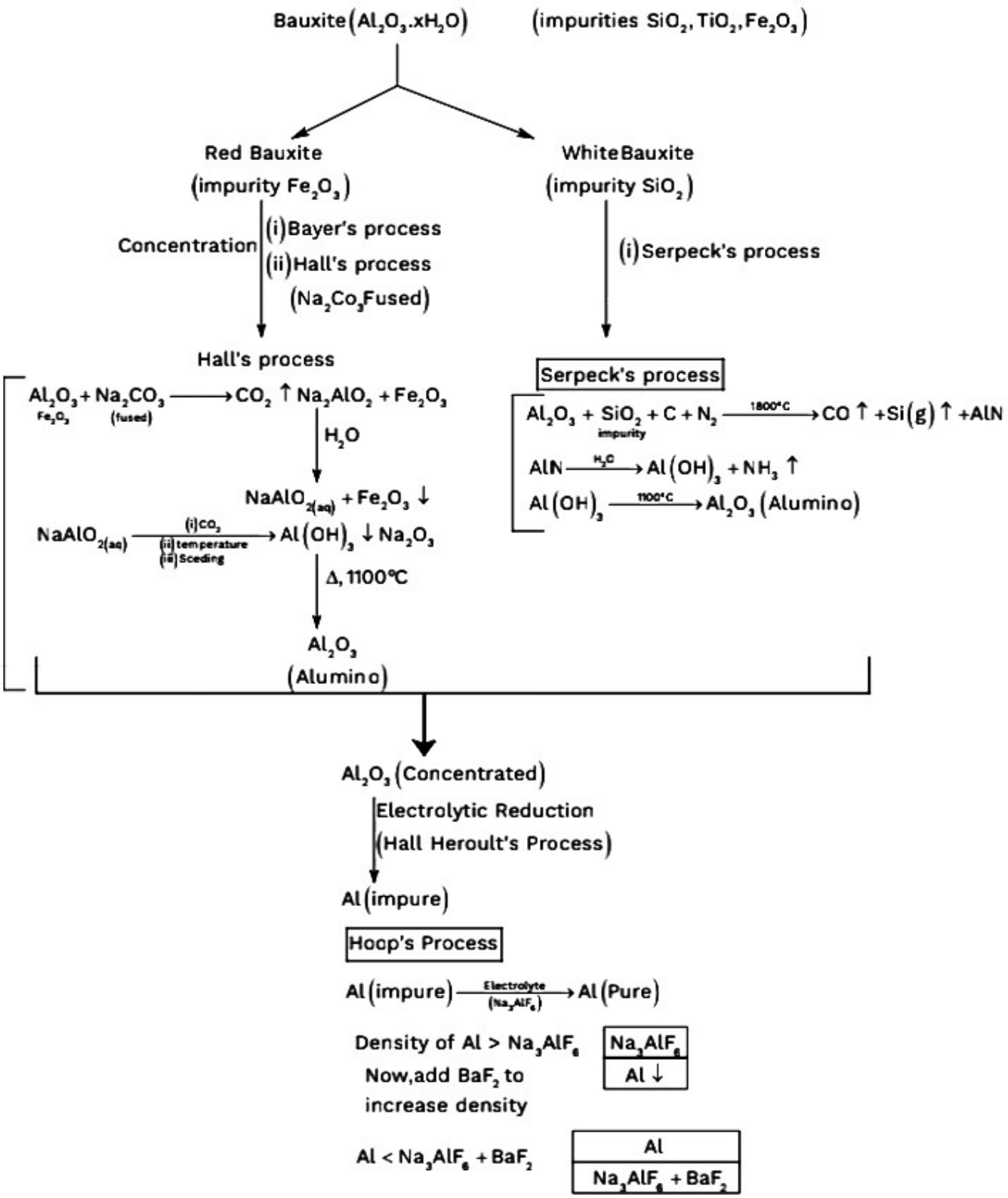
$Zn(g) + CO \uparrow$

↓ repid cooling

ZnS (Spelter)

(impure) ↓ refining → (i) Distilation, (ii) Electrorefining → (Al-Cathode) → Zn (Pure)

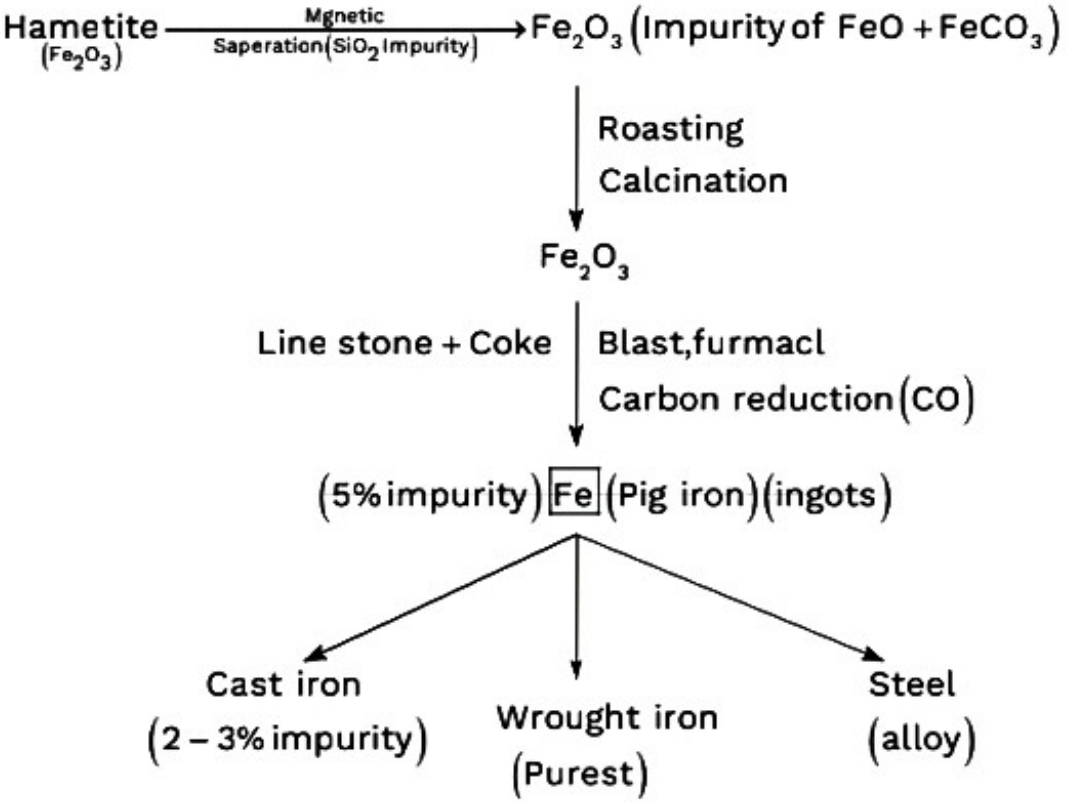
8. Aluminium (Al) :



$$\underbrace{Al\,impure + Cu + Si}_{Cathode} > Na_2AlF_6 + \underbrace{BaF_2 + Al\,epure}_{Cathode}$$

⇒ BaF_2, Cu & Si are used to increase density of respective layers.

9. Iron (Fe)



Recycling of iron :

$$\text{Pig iron} \xrightarrow[\text{Cupola furnace } (C+O_2)]{\text{Scrap iron}} \underset{(2\text{–}3\%\ \text{impurity})}{\text{Cast iron}}$$

Cast iron is formed by mixing pig iron & scrap iron with carbon in presence of 'air' & 'C' impurity oxidise by air.

Wrought iron :

Pig iron $\xrightarrow[Fe_2O_3(O.A.\,\&\,flux)]{Piddling\ process}$ wrought iron

Major imparity (5 % carbon)
Minor impurity (Si, P, S, Mn)
Impurities remove as – by taking Fe_2O_3 as O.A. & flux

$$Fe_2 + O_3 + \begin{bmatrix} C \rightarrow CO_2 \uparrow + Fe \\ S \rightarrow SO_2 \uparrow + Fe \\ P \rightarrow P_4O_{10} \xrightarrow{Fe_2O_3} FePO_4\ (\text{Fertilizer industry}) \\ Si \rightarrow SiO_2 \\ M_n \rightarrow M_nO \end{bmatrix} \Rightarrow \underset{(\text{Cement inductry})}{M_nSiO_3} + Fe$$

Steel :

Pig iron $\xrightarrow[\substack{(O.A + flux) \\ + \\ (\text{Element (x) disired})}]{Flux \Rightarrow CaO}$ Steel (Fe alloy)

Process		O.A. (Oxidising agent)	Remark
(1)	Bessemerisation	Hot air $(O_2 + N_2)$	Impurity of iron Nitrite introduced
(2)	Open hearth process	Hor air	Impurity of iron Nitrite introduced
(3)	Basic oxygen process	Hot O_2 pure	No impurity of Nitrite
(4)	L-D process (Electric are process) (Most efficient)	O_2Pure + Electrode	Impurity convert $C \rightarrow CO_2$ $S \rightarrow SO_2$ $\left.\begin{matrix} Si \rightarrow SiO_2 \\ M_n \rightarrow M_nO \end{matrix}\right] \rightarrow M_nSiO_3$

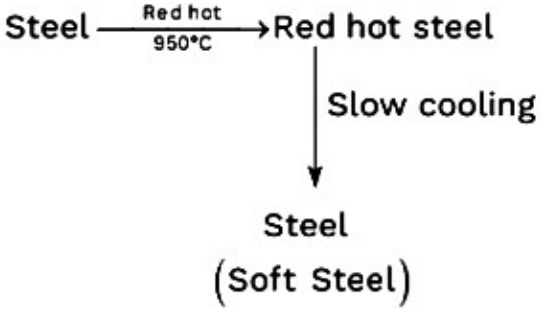
Element in steel	Property of steel
C, N	Hard steel
P, S	Brittle steel
Mn	High tensile strength (railway track)
Cr and Ni	Stainless steel (Oxide of Ce & Ni form protective layers)

Note

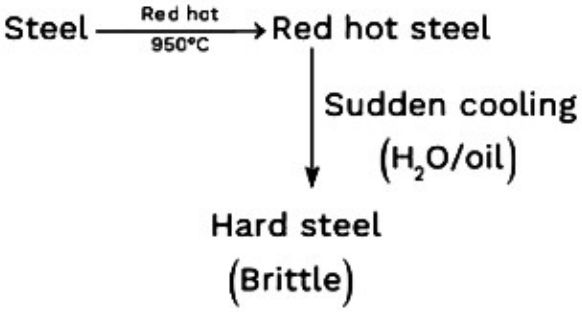
95% of Mn of world production is used to make steel, MnO_2 is mixed with Fe_2O_3 in blast furnace. Product obtained is called spiegeleisen (Fe + M_n + C) alloy.

Heat treatment of iron :

1. **Anealing ⇒**



2. **Quenching ⇒**



3. Tempering ⇒

Quenched steel $\xrightarrow[\text{(ii) Slow Cooling}]{\text{(i) 200°C}}$ Mild steel (Brittleness↓)

Case Hardering process ⇒ Formation of hard coating of iron carbide/Nitraide on Surface of Steel
- → Carbdising
- → Nitriding
- → Cyaniding

(i) Carbdising ⇒ Heating in presence of carbon

(ii) Nitriding ⇒ Heating in presence of N_2/NH_3

(iii) Cyaniding ⇒ Heating in presence of $K_4[Fe(CN)_6]$

10. Sodium (Na) ⇒ Downe cell process :

Pure NaCl (molten) electrolysis is not economical because

M.D. of NaCl = 803°C

B.P. of Na ≃ 880°C

So $CaCl_2$ is mixed with NaCl (Some KCl) to decrease melting point.

Working temperature now ≃ 600°C

So that Na-metal obtain in liquid state.

11. Potassium (k) :

$$KCl + Na \xrightarrow[\Delta]{885°C} K_{(g)} \uparrow + NaCl_{(2)}$$

(Reactivity Na > Reactivity of K)

Pure potassium 99.95% extracted.

12. Magnesium (Mg): Main ore of Mg is $KCl \cdot MgCl_2 \cdot 6H_2O$ dolomite

Process:

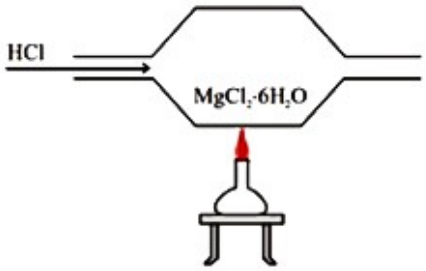
$$\text{Seawater} \xrightarrow{Ca(OH)_2} Mg(OH)_2 \downarrow$$

$$Mg(OH)_2 + HCl \longrightarrow MgCl_{2(aq.)}$$

$$MgCl_{2(aq)} \xrightarrow{\text{Crystalisution}} MgCl_2 \cdot 6H_2O$$

$$MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgO + HCl + H_2O$$

$$\xrightarrow{\text{dry HCl}} MgCl_{2(s)} + H_2O \uparrow$$



Electrolysis of $MgCl_2$

Electrolytic mixture $\Rightarrow MgCl_2 + CaCl_2 + NaCl$

$NaCl + CaCl_2$ used for decrease melting point and increases conductivity.

Reaction :

Cathode $\Rightarrow$ Steel

Anode $\Rightarrow$ Graphic

At Cathode $Mg^{+2} + 2e^- \longrightarrow Mg\,(Pure)$

At Anode $2Cl^- \longrightarrow Cl_2 \uparrow + 2e^-$

Some other process

(1) Chromatography

Column Chromatography $\Rightarrow$ In this method, two phases exist one is the mobile phase and the second in the stationary phase

Mixture of $\underset{\text{(Stationary phase)}}{Al_2O_3} \xrightarrow{\text{(mixture)}} \text{Mobile phase}$

Those with similar chemical properties are mixed in the mixture of Al_2O_3 and due to some small difference physical properties, they convert into mobile phase and Separated.

(2) Kroll's process $\Rightarrow$ For reduction of some metals.

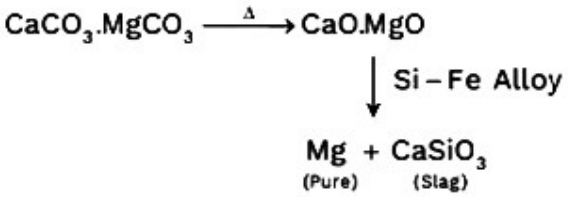
S-block metals are used as reducing agents in metallurgy.

$$TiCl_4 + Na / Mg \xrightarrow{\Delta} Ti + NaCl / mgCl_2$$

(3) Pidgeon process $\Rightarrow$

(Ferro silicon reduction of Mg)

Process :



Q110 **Which of following metal is extracted by thermal reduction process:**
(A) Al **(B) Fe** **(C) Cu** **(D) Mg**

Sol. **(B)**

Q111 **High Purity germanium is obtained by a technique that is based on:**
(A) Fractional distillation **(B) recrystallisation**
(C) Fractional crystallisation **(D) distillation**

Sol. **(C)**

Q112 **The metal that cannot be displaced form its aqueous Solution by zinc is-**
(A) Cu **(B) Ag** **(C) Al** **(D) Au**

Sol. **(C)**

Q113 **Stainless steel contain iron along with**
(A) Cr + Cu + C **(B) Ni + Zn + Cr** **(C) Cr + Ni + C** **(D) Ni + Cu + Mn**

Sol. **(C)**

Q114 The reaction $ZnO + C \longrightarrow Zn + CO$ is an example of-

(A) Combination reaction (B) Reduction-oxidation reaction

(C) Displacement Reaction (D) Decomposition Reaction

Sol. (C)

Q115 Among K, Mg, Au and Cu the one which is extracted by Heating its ore in air is-

(A) K (B) Mg (C) Au (D) Cu

Sol. (D)

Q116 Extraction of copper from copper pyrite $\left(CuFeS_2\right)$ involves:

(A) Crushing followed by concentration of the ore by froth flotation.

(B) removal of stag

(C) self-reduction step to step to product 'blister copper.'

(D) refining of 'blister copper by carbon reduction.

Sol. (ABC)

Q117 Match the anionic species given in column I & II with ore

Column (I) (ion)	Column (II) (ore)
(A) Carbonate	(P) Siderite
(B) Sulphide	(Q) Malachite
(C) Hydroxide	(R) Bauxite
(D) Oxide	(S) Calamite
	(T) Argentite

Sol. A→P,Q,S; B→T; C→R; D→R

Q118 Upon heating with Cu_2S the reagents that given copper metal is/are:-

(A) $CuFeS_2$ (B) CuO (C) Cu_2O (D) $CuSO_4$

Sol. (BCD)

Q119 The carbon-based reduction method is Not used of the extraction of

(A) Tin from SnO_2 (B) iron from Fe_2O_3

(C) Aluminium from Al_2O_3 (D) Mg from $MgCO_3, CaCO_3$

Sol. (CD)

Q120 Sulphide ore are common for the metals:

(A) Ag, Cu and Pb (B) Ag, Cu and Sn

(C) Ag, Mg and Pb (D) Al, C_4 and Pb

Sol. (A)